MR2006B · TECNOLÓGICO DE MONTERREY · AUTONOMOUS VEHICLE

# Vehicle Technical Reference

Hardware, physical modelling, control, state estimation, lateral aiding, path planning and communication of an STM32-controlled differential-drive robot for the Jardín de las Carreras circuit

| | |
|---|---|
| **Revision** | 1.0 — 14 October 2026 (course week 6 of 16) |
| **Platform** | NUCLEO-F446RE (STM32F446RET6, Cortex-M4F, 180 MHz), differential drive, 2 driven wheels + 2 casters |
| **Drivetrain** | 2 × JGB37-3530 / HS5141-178 (12 V, 178 rpm, 56.3:1, Hall encoder), 2 × BTS7960 half-bridge modules |
| **Status** | Drivetrain closed-loop on 50 Hz cascade, radio link running, bench and wheels-in-air identification complete; IMU, ground sensors, EKF, path follower and mission logic still ahead |
| **Sources** | `CONCEPT.md` Rev. F changeset, `BENCH_README.md`, `CALIB_README.md`, `MR2006B_CubeMX_Konfiguration.xlsx` (09.10.2026), challenge brief *Challenge MR2006B.pdf*, design sessions weeks 1–6 |
| **Supersedes** | *The Control System* overview (Rev. F, week 5) |

**How to read this document.** Every number is labelled by its origin. *Measured* values carry the date and set-up of the measurement; *catalogue* values are marked as such; *design* values are targets that have not been verified on hardware. Open work is tagged inline, using the same markers as `CONCEPT.md`:

`[MEASURE]` a quantity that must be measured `[VERIFY]` a value that exists but is not yet confirmed `[DECIDE]` a team decision `[ASK]` a question for the professor or a vendor `[RISK]` a known failure mode `[OPEN]` unimplemented.

All open items are collected in Section 14 with priority, owner action and the sections they block.

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## Summary

The vehicle is a ≈0.4 m long differential-drive robot with two driven wheels on a common axle and two passive casters. It must drive a closed circuit of roughly 150 m on paved paths around a garden of the Jardín de las Carreras, stop for 10 s at three waypoints, avoid obstacles, never touch grass, and return to its start — autonomously, powered by its own battery, and with an STM32 as the only computer on board. Speed is not graded; **completion, waypoint accuracy, obstacle handling and STM32-only operation are.**

The design follows from that rubric. Absolute positioning (RTK-GNSS) is excluded because a base station counts as an external aid. The vehicle therefore navigates by **dead reckoning that is continuously corrected by the environment**: wheel encoders give distance, a gyroscope gives heading, and downward- and sideways-looking time-of-flight sensors measure the grass/path boundary, benches and posts. A five-state Extended Kalman Filter fuses these. A cascade of controllers — mission logic, path follower, per-wheel position and speed control — turns the estimate into motion.

| Subsystem | State on 14 Oct 2026 |
|---|---|
| Drivetrain model | Linear model $u = K_v\omega + K_s$ confirmed on the bench ($R^2 > 0.999$); first-order plant identified on the BTS7960 ($K = 7.60\,(\mathrm{mm/s})/\%$, $\tau = 50$ ms) |
| Wheel control | Per-wheel cascade (trapezoid profile → P position → FF + PI speed) at 50 Hz; `D1000` ends at 999.5/1000.3 mm (firmware) |
| Odometry | Effective rolling diameter calibrated (72.6 mm with a common 3584 counts/rev); per-wheel scale and wheelbase still to be calibrated (UMBmark) |
| Communication | HC-12 at 9600 Bd running in parallel to the USB console |
| IMU | ICM-42688-P unobtainable; **Adafruit ISM330DHCX** selected, not yet integrated |
| Ground / boundary sensing | 5 × VL53L1X ordered: 4 planned on I2C1 (ground), 1 spare; not yet installed [OPEN] |
| EKF, path follower, mission | designed, not implemented [OPEN] |

The three facts that most shape the remaining work are: (i) the **entry heading error** at the start of a reference-free segment dominates the error budget (Section 12); (ii) the course itself has **not yet been measured**, which blocks wheel size, landmark map and error budget (Section 14); and (iii) the drivetrain is **not a matched pair** — the two motors differ by 12 % in speed and 20 % in breakaway duty at equal command, which the per-wheel feedforward and closed loop must and do absorb (Section 5.4).

# 1 Task and requirements

## 1.1 The challenge

The brief of course MR2006B (Sensors & Actuators / Microcontrollers) requires a vehicle capable of autonomous navigation. The binding requirements, taken verbatim in content from the brief, are:

- R1 The vehicle must be **four-wheeled**; steering may be Ackermann or differential.
- R2 Chassis, motors and drivers must be assembled by **week 5** and are reviewed by the professor.
- R3 At least **one control loop** must be implemented (e.g. speed or direction control) and must respond to a user-controlled reference.
- R4 The robot must **transfer data** (telemetry) to an external system.
- R5 It must operate in **Manual** mode (responds to speed/direction commands) and **Automatic** mode (follows a pre-programmed trajectory autonomously).
- R6 It must traverse the circuit and **return to the start**, stopping **10 s at each waypoint**; a circuit is valid only if all grass is avoided and the paths are used.
- R7 It must perform **simple obstacle avoidance**.
- R8 Time limit **15 min**; battery powered; good presentation (no dangling cables).

Figure 1 shows the circuit. Start and finish (X) lie next to the bell of the Jardín de las Carreras; the vehicle drives south-west to the first waypoint, north along the western path to the second waypoint, east along the northern path past a **step** (red) that runs alongside the path, to the third waypoint in the north-east and diagonally back to X. A **barrier** (black) borders the path south-east of X, next to a small building.




**Figure 1:** The circuit as given in the challenge brief. Left: satellite view with the course outline (yellow). Right: schematic with start/stop X, waypoints O, grass lines (green), step (red) and barrier (black). Driving direction is clockwise around the central green. Tree canopy covers most of the paths — one of the reasons GNSS was rejected even before the external-aid ruling. [MEASURE] No dimension of this course has been measured yet (Section 14.1).

## 1.2 Grading and what it implies

| **Final prototype criteria** | | | **Waypoint accuracy** | |
|---|---|---|---|---|
| Criterion | Points | Notes | Error [cm] | Grade |
| Complete the circuit alone | n/s | treated as all-or-nothing | 0–5 | 2.00 |
| No external aids to the sensors | n/s | excludes RTK base | 5–10 | 1.75 |
| Accuracy at waypoints | 2 | table right | 10–20 | 1.50 |
| Obstacle avoidance | 2/1/0 | reroute / wait / none | 20–40 | 1.25 |
| STM32 only | 2/1/0 | alone / with other MCU | 40–80 | 1.00 |
| Exposition | 10 | | 80–160 | 0.75 |
| | | | 160–320 | 0.50 |
| | | | 320–640 | 0.25 |
| | | | $> 640$ | 0 |

n/s = points not stated in the brief. The brief marks the accuracy column with ** but does not resolve the footnote: whether the error is taken per waypoint, as a mean or as the worst case is open [ASK].

The accuracy scale is **logarithmic**: every halving of the error buys 0.25 points. Going from 80 cm to 20 cm earns as much as going from 20 cm to 5 cm, but costs a fraction of the effort. Completion and STM32-only operation are binary. The design consequences:

**Design rules derived from the rubric.**

1. **Reliability beats precision.** A completed circuit with every waypoint in the 10 cm to 20 cm bracket (1.5 of 2 points) is the target. Chasing 0 cm to 5 cm is a 0.5-point gain at disproportionate risk.
2. **Speed is free.** Stop-and-turn instead of arcs, slow approach, 10 s ZUPT windows — all cost only time, of which there is a reserve (Section 9.3).
3. **Every correction is optional.** *No fix is the normal case, not a fault case.* The vehicle must complete the circuit on dead reckoning alone; each aiding source only improves it.
4. **Genuine rerouting**, not only waiting, earns the second obstacle point; the corridor geometry makes this a 1-D problem (Section 9.6).
5. **Only the STM32 decides.** Sensor modules with their own firmware (VL53L1X, HC-12) are peripherals; no co-processor runs navigation code. [ASK] The HC-12 contains an STM8 as its radio controller — to be confirmed with the professor as acceptable (Section 14.5, Q10).

### 1.3 Milestones

| Week | Milestone (graded) | Acceptance criterion used internally | State |
|---|---|---|---|
| 5 | Chassis, motor and driver built and working | drives straight and turns in place under command | done |
| 8 | Sensory system | IMU, encoders, ToF logged with timestamps, $< 1\,\%$ dropped frames | IMU chosen |
| 10 | Comms (telemetry) | full state logged at 10 Hz for 15 min | CSV link runs |
| 14 | Comms (remote control) | manual mode, failsafe verified | planned |
| 15 | Autonomous behaviour | full circuit without obstacles, waypoints within 40 cm | planned |
| 16 | Complete system | obstacles, waypoints within 20 cm, static obstacle evaded | planned |

Internal milestones in between: velocity loop tuned (week 8, done early on the 50 Hz cascade), EKF stationary and hand-pushed (week 10), UMBmark and Allan variance (week 12), path following on a 10 m × 10 m rectangle (week 13), full `-O2` competition build (week 13).

# 2 System overview

## 2.1 Architecture

The system is a cascade of four control layers with a state estimator beside them, all executed by one STM32F446RE without an operating system. Each layer generates the setpoint for the one below it; the estimator tells every layer where the vehicle is. Figure 2 shows the signals.

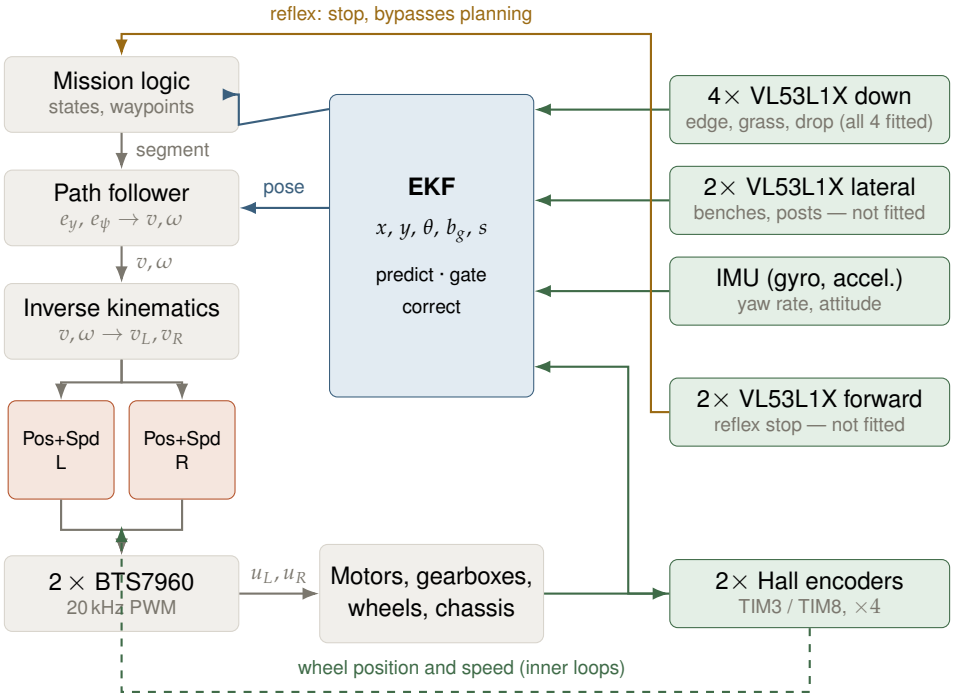



**Figure 2:** Signal flow. Setpoints flow down the left, measurements return on the right. The encoders feed two consumers: the wheel loops (fast, dashed) and the EKF (distance increment). Heading rate comes from the gyroscope, *not* from the encoder difference, because the latter depends on the effective wheelbase, which changes with every slip event. The forward sensors have a direct reflex path that stops the vehicle without passing through the planner.

## 2.2 Layers, rates and the present implementation

| Layer | Concept rate | Implemented | Content |
|---|---|---|---|
| 4 Mission logic | 20 Hz | — | state machine, waypoint sequencing, obstacle decisions |
| 3 Path follower | 100 Hz | — | cross-track/heading controller, TURN controller, speed profile |
| Estimator (EKF) | 200 Hz | — | prediction from encoder+gyro, corrections |
| 2 Wheel control | 200 Hz (PI) | **50 Hz** | trapezoid profile, P position, FF + PI speed, L/R sync |
| 1 Actuation | PWM | **20 kHz** | BTS7960 RPWM/LPWM, enable lines, slew limits |

**Concept versus implementation — deliberate deviation.** The concept places the wheel PI at 200 Hz in a 1 kHz tick. The firmware on the vehicle (teammate's `main.c`, extended 08–10 Oct) runs the whole wheel cascade at 50 Hz from TIM6. With velocity computed as a count difference, 50 Hz is the *correct* choice: one count per sample is 3.18 mm/s at 50 Hz but

12.7 mm/s at 200 Hz and 63.6 mm/s at 1 kHz (Section 4.3). Moving the loop faster requires a period-measurement (M/T) speed estimator first (Section 14.7). Rule of thumb for the cascade: at least a factor of 5 between adjacent loop bandwidths — this, not the sample rate, constrains the path follower (Section 6.6).

### 2.3 Design principles

These principles have decided every trade-off so far and are the argumentation basis for the final defence.

P1 **The rubric is the specification.** Speed optimisation and Ackermann steering were removed because they earn nothing.

P2 **Observability first.** Ask *which direction* a sensor constrains, not how well it sees (Section 7.3).

P3 **No fix is the normal case.** Every aiding measurement is optional and gated.

P4 **Prefer the observable failure.** A DC motor under overload slows down and the encoder sees it; a stepper loses steps silently. This rejected steppers, skid steer and GNSS fusion alike.

P5 **Timestamps beat sample rate.** A 200 Hz filter with exact $\Delta t$ beats a 1 kHz filter with jitter. Hence no RTOS, a 1 MHz free-running time base, and hardware input capture for data-ready edges.

P6 **Commission one layer at a time.** Open loop before closed loop, bench before vehicle, stationary EKF before driving EKF.

P7 **Measure the constant, do not inherit it.** $K_s$, $K_v$, gear ratio, wheel diameter, wheelbase are catalogue guesses until measured.

P8 **Verify the delivered part, not the listing.** The PH2.0 connector, the BTS7960 input threshold and the H-bridge truth table all differed from or were undocumented in their listings.

P9 **Structural safety over checks.** Forbidden driver states are made unreachable by construction (single setter), not tested for.

## 2.4 Rejected concepts

| Rejected | Reason |
|---|---|
| RTK-GNSS / GNSS+IMU fusion | base station ruled an external aid by the professor; canopy multipath; GNSS modules contain their own processors |
| SLAM | corridor degeneracy (scan matching unconstrained along the path axis); the only loop closure comes after all waypoints are scored; a static-world assumption violated by moving robots; a 10 cm grid of $40 \times 40$ m already needs 160 KB of the 128 KB SRAM |
| Line-scan camera | illumination under canopy; no gain over geometric (height/reflectance) boundary detection |
| Ackermann steering | no turn in place; larger turning envelope at the ramp; nothing graded rewards it |
| NEMA 17 steppers | silent step loss (unobservable failure), poor torque at speed, current draw at standstill |
| Ultrasonic sensors (HC-SR04) | silent false negatives on oblique surfaces — fatal in the reflex layer |
| Optical flow (PMW3901) | four independent grounds: (a) at 0.4 m/s, 100 fps and 150 mm height only 1.2 px of motion per frame (20–30 % quantisation, an order coarser than the encoders); (b) wind-driven tree shadow is indistinguishable from translation — a correlated phantom motion the EKF cannot reject; (c) it observes translation but not heading, the dominant error; (d) about two weeks of integration |
| Magnetometer heading | motor currents and structural steel distort the field ($\approx 8^\circ$ estimated near current-carrying wires) |
| FreeRTOS | context-switch jitter degrades $\Delta t$ in the EKF; harmonic cyclic executive suffices |
| TF-Luna LiDAR | cost; replaced by additional VL53L1X (one sensor type, one driver, no UART/DMA pressure) |
| C++ | ISR linkage pitfalls and hidden allocations; plain C with strict float discipline |

The BTS7960 itself was first rejected (its 74AHC244 input buffer has $V_{IH} = 0.7\,V_{CC}$, i.e. 3.5 V at 5 V supply), then adopted once it was recognised that powering the module's logic from 3.3 V lowers the threshold to $0.7 \times 3.3 = 2.31$ V, which the STM32's 3.3 V outputs satisfy.

# 3 Hardware

## 3.1 Bill of materials

Procurement is limited to local Mexican suppliers (Sandorobotics, MercadoLibre) plus DigiKey/-Mouser for parts without a local source. Table 1 lists every component on or for the vehicle, with its status.

**Table 1:** Complete hardware list. *Status:* installed, on hand, to order, decide, blocked.

| Item | Qty | Specification / role | Status |
|---|---|---|---|
| **Computing** | | | |
| NUCLEO-F446RE | 1 | STM32F446RET6, LQFP64, Cortex-M4F with single-precision FPU, 180 MHz, 512 KB flash, 128 KB SRAM; on-board ST-Link/V2-1 provides SWD, virtual COM port (USART2) and the 8 MHz MCO clock used as HSE | installed spare advisable |
| **Drivetrain** | | | |
| Gear motor JGB37-3530 / HS5141-178 | 2 | 12 V brushed DC, catalogue 178 rpm no-load, gear ratio 56.3:1, 3.4 kgf cm (rated or stall: [ASK]), stall current 4.3 A; 6 mm D-shaft | installed |
| Hall quadrature encoder | 2 | integrated on the motor shaft, 16 PPR per channel, $\times 4$ decoding $\Rightarrow$ 64 counts per motor revolution; open-collector outputs with $\approx 3\,\text{k}\Omega$ on-board pull-ups; supplied from 3.3 V | installed |
| BTS7960 module (IBT-2 / HW-039) | 2 | dual half-bridge (two BTS7960B), one module per motor; RPWM/LPWM inputs, R_EN+L_EN bridged; PWM up to 25 kHz; logic VCC on 3.3 V so that the 74AHC244 input threshold is $\approx 2.3$ V | installed |
| Previous driver (bench) | 1 | opto-isolated dual H-bridge, L298-style EN+IN1+IN2 logic, $\leq 10$ kHz PWM; used for the week-5 bench constants; truth table measured | retired |
| Hex adapter Pololu 2686 | 2 | 6 mm D-shaft to 12 mm hex, aluminium, 20 mm; keeps the wheel choice open | installed |
| Drive wheels | 2 | nominal 76 mm (initial code value); effective rolling diameter 72.6 mm calibrated with a common 3584 counts/rev | installed [VERIFY] |
| Casters | 2 | passive, size gated by the ramp-lip height | decide |
| Motor brackets | 2 | JGB37 brackets | installed |
| **Power** | | | |
| LiPo 3S 2200 mAh 35C | 1(+1) | 9.9 V to 12.6 V, 11.1 V nominal; 77 A short-circuit capability (fuse mandatory); second pack decided for 3-h sessions | 1 on hand 2nd to order |
| B3 balance charger | 1 | balance charging only via the 4-pin JST-XH lead | on hand |
| XT60 connector / pigtail, 14 AWG | 2 | battery disconnect point | to order |
| Fuse T10A (slow) + holder | 1 | main line; slow because a double stall draws 8.6 A | to order |
| Fuse F2–3A (fast) + holder | 1 | logic branch | to order |
| Main switch $\geq 10$ A | 1 | before the star point, breaks both branches | to order |
| Terminal block (star point) | 1 | only junction of motor and logic branches | to order |

| Item | Qty | Specification / role | Status |
|---|---|---|---|
| Electrolytic 470 µF to 1000 µF, 25 V | 1 | bulk capacitor at the drivers | to order |
| Ceramic 100 nF | 6 | 3 per motor (terminal–terminal, each terminal–case), soldered before mounting | kit on hand |
| Buck XY-3606, 5 V fixed | 1 | logic supply, Nucleo via E5V; fixed output (no trimpot) | installed [VERIFY] $U_{in,min}$ |
| LDO AMS1117-3.3 | 1–2 | separate 3.3 V sensor rail (keeps $V_{REF+}$ of the Nucleo clean) | installed overheated once |
| Battery divider 47 kΩ/10 kΩ | 1 | ratio 10/57 = 0.175 to PA0 (ADC1) | to fit |
| LiPo alarm, LiPo safety bag | 1+1 | independent cell monitor; charging safety | to order |
| Wire 14–16 AWG / 20 AWG / 20–22 AWG | — | battery–drivers / motor leads (crimped) / logic | partly |
| Standoffs | 1 set | driver underside can short | to order |
| **Sensing** | | | |
| VL53L1X ToF breakout | 5 | 4 fitted downward (ground/edge, I2C1), 1 spare; the 2 forward (reflex) and 2 lateral (benches, posts) positions are not covered [DECIDE]; individual XSHUT for address assignment | ordered |
| IMU (6-axis) | 1 | ICM-42688-P planned but unobtainable (0 stock, 16-week lead time); **Adafruit ISM330DHCX** selected; SPI + DMA, DRDY on input capture | to order/fit |
| **Communication** | | | |
| HC-12 radio (SI4463 based) | 2 | 433 MHz band, transparent UART at 9600 Bd, FU3, CH001; vehicle on USART1, ground station on FT232RL; SET on PC9 | running |
| FT232RL USB–UART adapter | 1 | ground station side (Mac) | on hand |
| **Development** | | | |
| Lab supply, multimeter | — | current-limited bring-up (6 V/100 mA truth-table test) | available |
| Oscilloscope | — | required for $U_{min}$ (driver output floor) and slot timing | access |
| Mac + STM32CubeIDE 2.2 + CubeMX | — | HAL-based C, Git in terminal | in use |

**Few spares.** No spare motor, driver, Nucleo or IMU is on hand (one spare VL53L1X). One Nucleo has already failed during the week-6 work (reported 10 Oct). A single defective part now costs a week of lead time in a 16-week schedule. [RISK] Recommended: one spare Nucleo, one spare BTS7960 module, one spare motor.

## 3.2 Microcontroller resources

### 3.2.1 Clock tree

The NUCLEO-F446RE has no crystal at X3. The 8 MHz comes from the ST-Link's MCO output, which is a clock *signal* and therefore needs HSE in **BYPASS** mode.

| Setting | Value | Reason |
|---|---|---|
| HSE | BYPASS, 8 MHz | MCO is a square-wave signal; PH1 stays free. Crystal mode "works" by overdriving the oscillator amplifier — marginal, not used |
| PLL source / M / N / P | HSE / 4 / 180 / 2 | VCO input 8/4 = 2 MHz (ST's recommendation for minimum jitter) → 360 MHz VCO → 180 MHz |
| AHB / APB1 / APB2 | /1 / /4 / /2 | HCLK 180, PCLK1 45 (timer clock **90**), PCLK2 90 (timer clock **180**) MHz |
| Voltage scale, over-drive, flash | VOS1, on, 5 WS | required at 180 MHz; set automatically by CubeMX |
| LSE / LSI | off / on | no RTC; LSI for the IWDG |
| CSS | off (target: on) | needs `HAL_RCC_CSSCallback` that pulls EN_L/EN_R low first [OPEN] |
| HSE start-up timeout | target 1000 ms | default 100 ms risks a failed start at power-on [VERIFY] |

**Why not HSI.** The internal RC oscillator is specified to ±1 % over 0 °C to 85 °C; the MCO derives from the ST-Link crystal (≈ ±30 ppm). A 1 % clock error is a *scale error on every timestamp* and therefore on every integrated quantity — speed, gyro integration, ZUPT averaging — and it is temperature-dependent. A PLLM error of /8 instead of /4 already halved every bus clock once (8 Oct) and made distance commands run at the wrong rate; the clock tree is therefore verified in generated code after every regeneration.

> **Battery operation without USB.** [RISK] HSE BYPASS depends on the ST-Link MCO. Whether the ST-Link section is powered and outputs MCO when the board is supplied through E5V only, without USB, is untested [VERIFY]. If it is not, the clock start fails and `Error_Handler` is entered. Mitigations in order: (1) test E5V-only start; (2) USB power bank on the ST-Link connector; (3) fit an 8 MHz crystal at X3 (with C33/C34 and solder bridges per UM1724) and switch HSE to crystal mode.

### 3.2.2 Timers

| Timer | Function | Clock | PSC | ARR | Result / note |
|---|---|---|---|---|---|
| TIM1 | PWM CH1–4 (BTS7960) | 180 MHz | 0 | 8999 | 20 kHz, set at run time by `PWM_Setup()`; 9000 steps = 0.011 % resolution |
| TIM3 | encoder L (PA6/PA7) | 90 MHz | 0 | 65535 | encoder mode TI1+TI2 (×4), IC filter 15, CKD /1 |
| TIM8 | encoder R (PC6/PC7) | 180 MHz | 0 | 65535 | as TIM3, CKD /2 to equalise the filter time |
| TIM6 | control tick | 90 MHz | 8999 | **199** | 50 Hz = 1/`CTRL_PERIOD_S`; last shared code had 119 ⇒ 83.3 Hz fix |
| TIM2 | 32-bit time base, IC on PA1 | 90 MHz | 0 → 89 | $2^{32}-1$ | target 1 MHz (wrap 71.6 min); IMU DRDY input capture [OPEN] |
| SysTick | HAL time base | HCLK | — | — | must stay SysTick; TIM6 as HAL time base would collide with the control callback |

**Encoder input filter.** Filter setting 15 samples 8 times at $f_{DTS}/32$. TIM3: $f_{DTS} = 90\,\text{MHz} \Rightarrow 8 \cdot 32/90\,\text{MHz} = 2.84\,\mu\text{s}$; TIM8 with CKD/2 also $f_{DTS} = 90\,\text{MHz} \Rightarrow 2.84\,\mu\text{s}$. Fastest edge spacing at no-load (Section 4.3) is 92 µs: the filter consumes 3 % of it. With the encoders' 3 kΩ pull-ups and an optional 1 kΩ/1 nF RC the rise time is ≈ 4 µs — still uncritical.

### 3.2.3 Pin allocation

| Pin | Morpho | Function | Target | Note |
|---|---|---|---|---|
| PA8 / PA9 | CN10-23 / -21 | TIM1_CH1/CH2 | BTS7960 L RPWM / LPWM | forward / reverse |
| PA10 / PA11 | CN10-33 / -14 | TIM1_CH3/CH4 | BTS7960 R RPWM / LPWM | |
| PB12 / PB13 | CN10-16 / -30 | GPIO out, pull-down | EN_L / EN_R (R_EN+L_EN) | plus external 10 kΩ to GND: defined off during reset |
| PA6 / PA7 | CN10-13 / -15 | TIM3_CH1/CH2 | encoder L A / B | |
| PC6 / PC7 | CN10-4 / -19 | TIM8_CH1/CH2 | encoder R A / B | TIM4 impossible: would remove USART1 |
| PA2 / PA3 | internal | USART2 | ST-Link VCP | 115200 Bd console |
| PB6 / PB7 | CN10-17 / CN7-21 | USART1 | HC-12 RXD / TXD | assign PB6/PB7 first, else CubeMX picks PA9/PA10; PB7 pull-up [VERIFY] |
| PC9 | CN10-1 | GPIO out | HC-12 SET | high = transparent, low = AT |
| PA13 / PA14 | CN7-13 / -15 | SWD | ST-Link | debug = Serial Wire (not trace) |
| PH0 | CN7-29 | RCC_OSC_IN | MCO 8 MHz | PH1 free in BYPASS [VERIFY] |
| PA1 | CN7-30 | TIM2_CH2 IC | IMU INT1/DRDY | reserved |
| PB3/PB4/PB5, PA4 | — | SPI1 + CS | IMU | relocated from PA5–PA7 (taken by TIM3); costs SWO |
| PB8 / PB9 | — | I2C1 | 4 ground VL53L1X | reserved |
| PB10 / PC12 | — | I2C2 | (forward/lateral VL53L1X, not fitted) | PC12 as I2C2_SDA [VERIFY] in CubeMX |
| PA0 | — | ADC1_IN0 | battery divider | PC4/PC5 further ADC channels reserved |
| 4 GPIO | — | XSHUT | VL53L1X (ground) | pins per `MR2006B_Pinout.xlsx` [VERIFY] |
| PA5 / PC13 | — | LD2 / B1 | on-board | unused |

Three constraints only appear late in CubeMX and are therefore fixed here: (a) the second encoder must use TIM8, because TIM4's only encoder pins PB6/PB7 are USART1's only alternative to PA9/PA10, which TIM1 occupies; (b) SPI1 must use PB3/PB4/PB5 because PA6/PA7 carry TIM3; (c) USART3 must use PC10/PC11 because PB11 does not exist in LQFP64. **EXTI lines are shared by pin number**, not by port: interrupt inputs must use distinct pin numbers.

### 3.2.4 Interrupt priorities and DMA

NVIC priority group 4 (4 bits pre-emption, 0 sub-priority).

| Source | Pre-emption | State | Note |
|---|---|---|---|
| SysTick | 0 | on | `HAL_GetTick` must advance even inside ISRs |
| TIM6 | 1 | on | control tick |
| USART1, USART2 | 3 | on | HC-12, console; without NVIC the console falls silent |
| TIM2, EXTI15:10 | — | [VERIFY] | present in `.ioc`, unused |

**Planned DMA allocation** (stage with IMU and ADC, [VERIFY] against RM0390 Table 28/29 before wiring): SPI1 RX/TX on DMA2 streams 0/3, ADC1 on DMA2 stream 4 (circular, IRQ disabled), USART1 TX on DMA2 stream 7 and RX on DMA2 stream 2 (circular, IDLE-line interrupt). USART2 stays without DMA so that DMA1 streams 5/6 remain free for I²C. Dropping the TF-Luna removed three to four UART DMA streams and with them the DMA2 bottleneck.

### 3.3 Power architecture

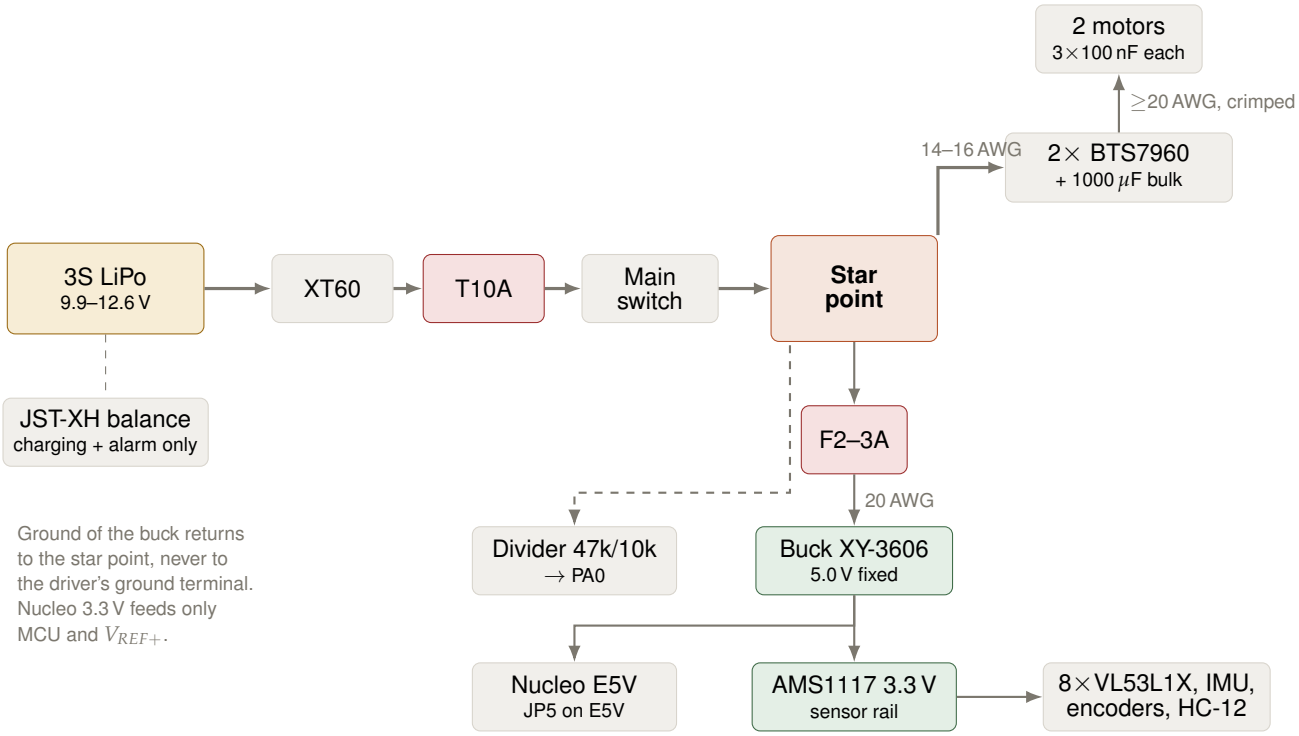



**Figure 3:** Power topology. The star point is the only place where motor and logic currents meet; the buck converter's return goes there, not to the driver, so that motor ripple does not travel through shared wire impedance into the sensor supply. The balance lead is never a supply: drawing from it discharges single cells and creates the drift that balance charging exists to prevent.

**Ratings.** A 35C, 2.2 Ah pack can deliver $35 \times 2.2\,\text{A} = 77\,\text{A}$ into a short circuit; the main fuse is the only part between a short and that current. It is slow-blow because a simultaneous stall of both motors draws $2 \times 4.3\,\text{A} = 8.6\,\text{A}$. Dupont contacts ($\approx 1\,\text{A}$) are acceptable only for encoder and logic lines; motor leads are crimped or soldered, because an oxidised contact adds a drifting series resistance that the feedforward cannot see (Section 4.4).

**Battery measurement.** $U_{PA0} = U_{bat} \cdot 10/57$: 2.21 V at 12.6 V, 1.74 V at 9.9 V, both below the 3.3 V ADC range; one LSB corresponds to 4.6 mV of pack voltage. Without the divider PA0 floats and reads 100 mV to 240 mV of noise. [MEASURE] actual ratio if 5 % resistors are used.

**Thresholds.** Software cut-off at 10.0 V (3.33 V/cell). [VERIFY] The buck's minimum input voltage must lie below the worst case at its input: $9.9\,\text{V} - 0.34\,\text{V}\ (\text{stall sag}) - 0.1\,\text{V}\ (\text{wiring}) \approx 9.4\,\text{V}$; otherwise the logic browns out *before* the software cut-off acts, which presents as a random reset.

**3.3 V budget.** Worst case (originally eight VL53L1X ranging, HC-12 transmitting at full power; only four are fitted, so this is conservative) $\approx 0.46\,\text{A}$; the AMS1117 then dissipates $(5 - 3.3)\,\text{V} \times 0.46\,\text{A} \approx 0.78\,\text{W}$ — hot on a small SOT-223 module. The observed overheating with the HC-12 attached (which also pulled down the Nucleo's 3.3 V and disturbed the ST-Link) is open item O-4. [DECIDE] Feed the HC-12 from 5 V (its supply range allows it; verify that its TXD level stays STM32-compatible), or split the sensor rail across two LDOs.

**Commissioning order of the harness.** (1) build complete without battery or loads; (2) continuity: XT60+ to star point, XT60− to star point, no path between them; (3) battery in, switch on, check polarity and voltage at the star point; (4) connect the buck, measure (5.00 ± 0.10) V *unloaded*; (5) only then the Nucleo on E5V; (6) drivers and motors last. Step 4 before step 5 is the one that saves the Nucleo from a defective converter.

**Session energy.** [MEASURE] Mean pack current while driving. With an assumed 1 A mean the 2.2 Ah pack lasts $\approx$ 1.5 h to 70 % depth of discharge; the 15-min run is uncritical, a 3-h test afternoon is not — hence the second pack.

# 4 Physical modelling

This section derives every model the firmware uses or will use, states its assumptions, and identifies which parameters are measured and which are not.

## 4.1 Frames and notation

| Symbol | Meaning |
|---|---|
| $\{W\}$, $\{B\}$ | world frame (horizontal plane, $z_W$ up; origin at X, $x_W$ along the initial heading) and body frame ($x_B$ forward, $y_B$ left, $z_B$ up, origin at the drive-axle midpoint) |
| $x$, $y$, $\theta$ | position of the axle midpoint and heading (yaw) in $\{W\}$; angles counter-clockwise positive |
| $v$, $\omega$ | forward speed and yaw rate of the body |
| $v_L$, $v_R$; $\omega_L$, $\omega_R$ | wheel ground speeds; wheel angular speeds |
| $r = D/2$, $b$ | effective rolling radius; effective wheelbase (track width), 223.25 mm geometric |
| $i$, $\eta$ | gear ratio (motor : output), gearbox efficiency |
| $C$, $c = \pi D/C$ | encoder counts per wheel revolution; distance per count |
| $d \in [-1, 1]$, $u = 100\,d$ | PWM duty (signed), in % where so stated |
| $U_{bat}$ | battery voltage at the driver input |
| $T_s$ | sample period of the wheel loop (20 ms) |
| $b_g$ | gyroscope bias; $s$ odometry scale state in the EKF (unitless, $\approx 1$) |

## 4.2 Differential-drive kinematics

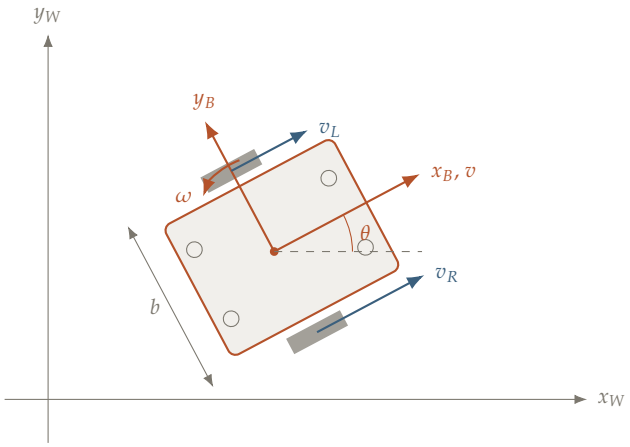



**Figure 4:** Differential drive. The driven wheels sit on one axle through the body origin; two casters carry the rest of the load. With $v_R > v_L$ the vehicle turns left (positive $\omega$).

Rolling without slip gives $v_{L,R} = r\,\omega_{L,R}$. Forward and inverse kinematics of the axle midpoint:

$$v = \tfrac{1}{2}\,(v_R + v_L), \qquad \omega = \frac{v_R - v_L}{b}, \tag{1}$$

$$v_R = v + \tfrac{b}{2}\,\omega, \qquad v_L = v - \tfrac{b}{2}\,\omega. \tag{2}$$

The instantaneous centre of rotation (ICR) lies on the axle line at signed distance $R = v/\omega = \frac{b}{2}\,(v_R + v_L)/(v_R - v_L)$; for $v_R = -v_L$ it coincides with the axle midpoint (turn in place), which is the reason the IMU is to be mounted there. The motion equations (unicycle model) are

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega, \tag{3}$$

subject to the **non-holonomic constraint** (NHC) that the body cannot move sideways:

$$-\dot{x}\sin\theta + \dot{y}\cos\theta = 0. \tag{4}$$

**Discrete odometry.** From count increments $\Delta n_{L,R}$ in one sample, $\Delta s_{L,R} = c_{L,R}\,\Delta n_{L,R}$, $\Delta s = \frac{1}{2}(\Delta s_R + \Delta s_L)$, $\Delta\theta = (\Delta s_R - \Delta s_L)/b$. The exact update for a constant-curvature step is

$$\begin{aligned} x_{k+1} &= x_k + \Delta s\ \mathrm{sinc}\!\left(\tfrac{\Delta\theta}{2}\right)\cos\!\left(\theta_k + \tfrac{\Delta\theta}{2}\right), \\ y_{k+1} &= y_k + \Delta s\ \mathrm{sinc}\!\left(\tfrac{\Delta\theta}{2}\right)\sin\!\left(\theta_k + \tfrac{\Delta\theta}{2}\right), \qquad \theta_{k+1} = \theta_k + \Delta\theta, \end{aligned} \tag{5}$$

with $\mathrm{sinc}(z) = \sin z/z$. Dropping the sinc factor (midpoint rule) causes a relative chord error of $\Delta\theta^2/24$; for the fastest planned spin ($\pm 300$ mm/s per wheel, $\omega = 2.69$ rad/s, $\Delta\theta = 0.054$ rad at 50 Hz) that is $1.2 \times 10^{-4}$ — negligible, so the EKF uses the midpoint rule. In the EKF, $\Delta\theta$ is taken from the gyroscope, not from the wheels (Section 7).

**Turning envelope.** For the concept envelope of 400 mm × 350 mm with the axle at mid-length, a turn in place sweeps a circle of diameter $2\sqrt{0.2^2 + 0.175^2}$ m $= 0.53$ m; 0.6 m of free width is planned. [VERIFY] final chassis dimensions.

### 4.3 Encoder and wheel-position measurement

**Resolution.** The Hall encoder sits on the motor shaft, ahead of the gearbox: 16 pulses per channel and revolution, decoded on both edges of both channels (×4), gives 64 counts per motor revolution and

$$C = 64\,i \quad \text{counts per wheel revolution.} \tag{6}$$

| | Motor L | Motor R | Catalogue |
|---|---|---|---|
| Counts per output revolution (hand-turned, mean of 3) | 3595 | 3618 | 3603 |
| Implied gear ratio $C/64$ | 56.17 | 56.53 | 56.3 |
| Rear (motor) shaft, one revolution | 64 | 64 | 64 |

**Interpretation of the hand-turn counts.** Two gearboxes of the same type have identical tooth counts and therefore identical ratios. A difference of 23 counts/rev is 2.3° of output rotation — within reach of a felt-pen mark plus gearbox play when turning by hand. The 0.64 % difference is therefore *not* accepted as a gear-ratio difference. What matters for odometry is the **effective distance per count per wheel**, which includes the rolling radius of each tyre and is only measurable by driving (UMBmark, Section 13.2). A finer static check: wheel mounted, fixed pointer at the tyre, 10 revolutions always approaching the mark from the same direction (resolves $\approx \pm 1$ count/rev).

The vehicle firmware currently uses a common $C = 3584$ ($= 64 \times 56$) together with an effective diameter $D = 72.6$ mm calibrated from a straight run (firmware 1000 mm, tape 955 mm at $D = 76$ mm). Only the ratio $D/C$ is calibrated:

$$c = \frac{\pi \cdot 72.6\,\mathrm{mm}}{3584} = 0.063\,64\,\mathrm{mm/count}. \tag{7}$$

Switching to per-wheel counts requires raising $D$ to $72.6 \cdot 3606.5/3584 \approx 73.06$ mm to keep the mean scale.

**Edge rate and input filter.** At the no-load output speed of 181 rpm the motor turns at $181 \times 56.17 = 10\,167$ rpm, giving $10167/60 \times 64 = 10.8$ kedges/s, i.e. 92 µs between edges. The 2.84 µs digital filter (Section 3.2.2) is 3 % of that.

**Speed quantisation.** With count differencing over one sample,

$$\hat{v}_{raw,k} = \frac{c\,\Delta n_k}{T_s}, \qquad \delta v = \frac{c}{T_s} = \begin{cases} 3.18\,\text{mm/s} & T_s = 20\,\text{ms}\ (50\,\text{Hz}) \\ 12.7\,\text{mm/s} & T_s = 5\,\text{ms}\ (200\,\text{Hz}) \\ 63.6\,\text{mm/s} & T_s = 1\,\text{ms}\ (1\,\text{kHz}) \end{cases} \tag{8}$$

The firmware smooths with a first-order IIR, $\hat{v}_k = \hat{v}_{k-1} + \alpha(\hat{v}_{raw,k} - \hat{v}_{k-1})$, $\alpha = 0.5$, equivalent time constant $\tau_f = -T_s/\ln(1-\alpha) = 28.9$ ms — of the same order as the plant time constant (50 ms) and therefore part of the loop's effective dead time (Section 6.3). Differences are formed in `int16_t`, which makes the 16-bit counter wrap-around harmless as long as $|\Delta n| < 32768$ per sample.

**Backlash.** Because the encoder sits before the gearbox, every reversal turns the motor through the gear play before the wheel moves; those counts enter odometry as distance that did not happen. The lost motion $\delta_b$ (counts) is measured per wheel by calibration test `L` **[MEASURE]**. Consequence for control: **every positioning move ends from one direction**, critically damped, without overshoot-and-return.

### 4.4 DC motor, gearbox and driver

**Equations.** Per motor (armature resistance $R_a$, inductance $L_a$, back-EMF constant $k_e$, torque constant $k_t$ — numerically equal in SI units), referred to the motor shaft:

$$L_a \frac{\mathrm{d}i_a}{\mathrm{d}t} = U_a - R_a i_a - k_e\,\omega_m, \tag{9}$$

$$J_{eq} \frac{\mathrm{d}\omega_m}{\mathrm{d}t} = k_t\,i_a - \tau_f(\omega_m) - \frac{\tau_w}{i\,\eta}, \tag{10}$$

$$J_{eq} = J_m + J_{gb} + \frac{J_w + \frac{m}{2}r^2}{i^2\,\eta}, \qquad \omega_m = i\,\omega_w, \tag{11}$$

where $\tau_w$ is the load torque at the wheel (rolling resistance, grade, acceleration of the vehicle's share is already in $J_{eq}$) and $\frac{m}{2}r^2$ is half the vehicle's translational inertia per wheel for straight driving.

**PWM averaging.** The BTS7960 is driven sign–magnitude: one half-bridge receives PWM (RPWM or LPWM), the other stays low, both enables high. During the off-time both low-side switches conduct and the winding is short-circuited (synchronous slow decay). As long as the current does not reach zero within a PWM period (*continuous conduction*, $L_a/R_a \gg 50$ µs), the winding sees the average $U_a = d\,U_{bat}$ and the duty-to-speed relation is linear. The previous driver PWM'd the enable line, so the off-state was coast (diode recirculation); at low duty this produces discontinuous conduction and a non-linear low end. **[MEASURE]** $L_a$ (LCR meter or current step) to confirm continuous conduction at 20 kHz.

**Steady state.** With $\mathrm{d}i/\mathrm{d}t = \mathrm{d}\omega/\mathrm{d}t = 0$, friction $\tau_f = \tau_C \operatorname{sgn}\omega_m + b_v\omega_m$ and a driver voltage drop $U_{dr}$, eliminating $i_a$ and solving for the duty gives, for wheel speed $n$ in rpm,

$$\boxed{d = K_v\, n + K_s \operatorname{sgn} n} \quad \text{with} \quad K_v = \frac{2\pi\, i}{60\, U_{bat}}\left(k_e + \frac{R_a b_v}{k_t}\right), \quad K_s = \frac{1}{U_{bat}}\left(\frac{R_a}{k_t}\left(\tau_C + \frac{\tau_w}{i\eta}\right) + U_{dr}\right). \tag{12}$$

This is the model behind the feedforward, and it explains four observations:

1. **Both constants scale with $1/U_{bat}$.** Between full (12.6 V) and empty (9.9 V) the required duty for the same speed changes by 27 %. Hence the feedforward is to be normalised, $d_{ff} = (K_v^V n + K_s^V)/U_{bat}$, with constants in volts identified together with $U_{bat}$ (calibration `U` command) and $U_{bat}$ measured on PA0 [OPEN].
2. **$K_s$ contains the load.** Rolling resistance, grade and caster friction enter through $\tau_w$; a bench (no-load) $K_s$ is a lower bound for driving. Ground identification (`G`) gives the values for the vehicle.
3. **$K_s$ contains the driver.** A bipolar bridge drops volts, a MOSFET bridge (BTS7960: tens of mΩ against an armature of $\approx 2.8\,\Omega$) practically nothing. The BTS7960 dead zone (7.0 % mean, wheels in air) is indeed lower than the bench intercepts (11.0/12.7 %).
4. **$K_v$ is a motor property.** It should not change with the driver. Bench: $1/K_v$ corresponds to 7.73 (mm/s)/% (L) and 7.50 (mm/s)/% (R); BTS7960 identification: 7.60 (mm/s)/% mean. Agreement to 2 % — a consistency check that the identification is sound (differences in $U_{bat}$ between the two sessions are not recorded [VERIFY]).

**Dynamics.** Neglecting $L_a$ (electrical time constant ≪ mechanical), Eqs. (9) and (10) reduce to a first-order plant from duty to speed,

$$G(s) = \frac{V(s)}{U(s)} = \frac{K}{\tau_m s + 1}, \qquad \tau_m = \frac{J_{eq} R_a}{k_e k_t + R_a b_v}, \tag{13}$$

identified on the BTS7960 with wheels in the air as $K = 7.596\,(\mathrm{mm/s})/\%$, $\tau_m = 50\,\mathrm{ms}$, no measurable dead time. On the ground $J_{eq}$ grows by $\frac{m}{2}r^2/(i^2\eta)$; with an assumed $m = 3\,\mathrm{kg}$ that is $\approx 6\times10^{-7}\,\mathrm{kg\,m^2}/\eta$ at the motor shaft — comparable to the rotor inertia of a 37 mm motor, so $\tau_m$ on the ground may be noticeably larger [MEASURE] (test `G`, step response on the floor).

**Parameter estimate (not measured).** Assuming the bench supply was 12 V: at $d = 1$ the model gives 181 rpm, of which the fraction $K_s = 0.11$ of the voltage covers friction, so $k_e \approx (1 - K_s)U/\omega_m = 0.89 \cdot 12/1065\,\mathrm{rad/s} \approx 10.0\,\mathrm{mV\,s/rad}$. With the catalogue stall current, $R_a \approx 12\,\mathrm{V}/4.3\,\mathrm{A} = 2.8\,\Omega$ and the stall torque at the motor shaft $\approx k_t \cdot 4.3\,\mathrm{A} = 43\,\mathrm{mN\,m}$, i.e. $\approx 2.4\,\mathrm{N\,m}\cdot\eta$ at the output. Even with a poor $\eta$ this is several times the catalogue 3.4 kgf cm (0.33 N m), which therefore is almost certainly the *rated* and not the stall torque. [VERIFY] bench voltage, and [ASK] vendor.

### 4.5 Friction

The drivetrain friction is modelled as Coulomb + viscous + Stribeck:

$$\tau_f(\omega) = \left(\tau_C + (\tau_S - \tau_C)\, e^{-(\omega/\omega_S)^2}\right) \operatorname{sgn}\omega + b_v\,\omega. \tag{14}$$

Static friction $\tau_S$ exceeds kinetic friction $\tau_C$. On the bench this is directly visible:

| | Breakaway from rest | Running intercept $K_s$ | Difference | Ratio |
|---|---|---|---|---|
| Motor L | 0.141 | 0.110 | 0.031 | 1.28 |
| Motor R | 0.169 | 0.127 | 0.042 | 1.33 |

**Consequences.** (i) The *running* intercept belongs in the speed feedforward; the breakaway value would overcompensate every normal movement. (ii) Starting from rest needs more — either the integrator winds up through the gap (slow, produces stick-slip with a brief slip of the wheel) or the feedforward switches to the breakaway value below a speed threshold (variant B, Section 6.3). (iii) The breakaway scatters by $\approx \pm 5\,\%$ with shaft angle (gear meshing), so it is measured as a distribution, not a value (test `B`: six trials per direction, maximum used).

### 4.6 Longitudinal vehicle dynamics and motor margin

On a grade $\alpha$ with rolling-resistance coefficient $c_{rr}$, the traction force both wheels must deliver is

$$F = m\big(g(\sin\alpha + c_{rr}\cos\alpha) + a\big), \qquad \tau_w = \tfrac{1}{2}Fr \text{ per wheel.} \tag{15}$$

Evaluated with assumed $m = 3\,\text{kg}$ [MEASURE], $c_{rr} = 0.02$ (hard tyre on paving, assumption), $r = 36.3\,\text{mm}$:

| Grade | Accel. | $F$ [N] | $\tau_w$ per wheel [N m] | [kgf cm] | share of 3.4 kgf cm |
|---|---|---|---|---|---|
| 0° | 0 | 0.59 | 0.011 | 0.11 | 3 % |
| 0° | $0.3\,\text{m/s}^2$ | 1.49 | 0.027 | 0.28 | 8 % |
| 5° | 0 | 3.15 | 0.057 | 0.58 | 17 % |
| 8° | $0.3\,\text{m/s}^2$ | 5.58 | 0.101 | 1.03 | 30 % |

Torque is not the constraint even if 3.4 kgf cm were the stall rating. The binding limits are elsewhere:

- **Wheel diameter versus edges.** The lip at the ramp must be crossed by wheels and casters. (The step marked on the northern path in Fig. 1 runs *parallel* to the direction of travel: it is a drop-off to stay clear of and a lateral reference, not an edge to climb [VERIFY] on site.) A free (caster) wheel of radius $r_c$ carrying normal load $N$ and pushed horizontally against an edge of height $h < r_c$ starts to climb only when the push exceeds

$$F_{push} = N\tan\beta = N\,\frac{\sqrt{h\,(2r_c - h)}}{r_c - h}, \qquad \cos\beta = \frac{r_c - h}{r_c}. \tag{16}$$

  Example: $r_c = 25\,\text{mm}$, $h = 10\,\text{mm}$ gives $F_{push} = 1.33\,N$ — the casters can need more force than the whole vehicle needs on an 8° ramp. For a driven wheel the same geometry applies with the friction limit $\mu N$ in place of the push. [MEASURE] ramp-lip height — this gates wheel and caster size.

- **Traction.** The drive force is limited by $\mu N_d$, where $N_d$ is the normal force on the drive wheels. With the axle at mid-length and casters fore and aft, $N_d < mg$; at a crest or in a dip the casters can unload the drive wheels. [VERIFY] load split on the finished chassis.

- **Speed.** Bench no-load maxima 688 mm/s (L) and 673 mm/s (R) at $d = 1$; BTS7960, wheels in air: highest measured 680 mm/s. A commanded 800 mm/s (`D2000` test, 8 Oct) saturated and fell short (1766/1759 mm). Recommended ceiling: 75 % of the weaker wheel at 10.5 V: $654.5 \times 10.5/12 \times 0.75 \approx 430\,\text{mm/s}$ — `V_NOM_MAX` should drop from 800 to 450–500 (open item O-5).

- **Minimum speed.** The weaker wheel's breakaway limits slow creeping; estimated $\approx 30\,\text{mm/s}$ [VERIFY] with test `B` on the BTS7960.

### 4.7 Rotation, casters and effective wheelbase

Yaw dynamics, with $I_z$ the yaw inertia and $M_c$ the caster and tyre-scrub moments,

$$I_z\,\dot{\omega} = \tfrac{b}{2}\,(F_R - F_L) - M_c. \tag{17}$$

Swivel casters produce a transient moment whenever the direction of travel changes (they must flip), which disturbs the first centimetres after a turn or reversal. Procedure consequence: after a turn in place drive 20 cm to 30 cm straight before re-acquiring references. The tyre contact patches have finite width, so the *effective* wheelbase differs from the geometric 223.25 mm; it is identified by UMBmark (factor $E_b$) and by 10-turn spins (`W±3600`).

### 4.8 Odometry error model

**Systematic errors** (repeatable, calibratable):

- **Scale error** $\varepsilon_s$ (mean $D/C$ wrong): along-track error $\delta s = \varepsilon_s\, s$.
- **Unequal wheels** $E_d = D_R/D_L \neq 1$, or $\varepsilon = 1 - E_d$: driving equal counts makes the vehicle turn with curvature $\kappa = \varepsilon/b$, so the lateral deviation grows **quadratically**,

$$y(s) = \frac{\varepsilon\, s^2}{2b}. \tag{18}$$

- **Wheelbase error** $E_b$: wrong encoder-derived heading change in turns, proportional to the angle turned.
- **Backlash** $\delta_b$: lost motion per reversal.

With the hand-turn values ($\varepsilon = 0.64\,\%$, $b = 0.223\,\text{m}$) and a common count constant, Eq. (18) predicts a drift to the right of 14 mm after 1 m, 57 mm after 2 m and 36 cm after 5 m (curvature $0.0287\,\text{m}^{-1}$, 1.6° per metre) — the direction actually observed on the vehicle.

> **Telling causes apart by the shape of the drift.** Drive `D1000`, `D2000`, `D4000` and measure the lateral offset $y$: $y \propto s^2 \Rightarrow$ scale difference between the wheels ($E_d$, fixed by UMBmark); $y \propto s \Rightarrow$ constant heading error from the start (placement, backlash, start transient); $y \approx$ const. after the first metre $\Rightarrow$ L/R tracking difference during the ramp-up (fixed by the synchronisation term, Section 6.5).

**Non-systematic errors** (random): wheel slip on dust, gravel or wet paving, uneven ground, the step, caster disturbances. They cannot be calibrated, only detected: the **slip detector** compares the encoder yaw rate $(\Delta s_R - \Delta s_L)/(b\,T_s)$ with the bias-corrected gyro rate; a persistent difference inflates the process noise of the distance increment in the EKF (Section 7). With the gyro providing heading, $E_d$ and $E_b$ no longer corrupt the EKF's heading — but they still corrupt the encoder-only firmware in use today, the slip detector and the L/R synchronisation, which is why UMBmark remains worth doing.

### 4.9 Inertial model and slope projection

The gyroscope measures the body-frame rate with bias, noise and a temperature dependent offset,

$$\omega_{z,meas} = \omega_z + b_g(T,t) + n_\omega, \qquad \dot{b}_g = n_b \ \text{(random walk)}. \tag{19}$$

Heading integrated from a residual bias $\delta b$ produces, at speed $v$ over distance $s$, a lateral error $y = \delta b\, s^2/(2v)$; angle random walk $N$ [°/$\sqrt{\text{h}}$] adds $\sigma_\theta(t) = N\sqrt{t}$. Both are determined from an Allan-variance run of the selected IMU **[MEASURE]**.

During every stop the true rate is zero, so the mean of the gyro output over the stop window is a direct bias measurement (ZUPT). For white rate noise of standard deviation $\sigma_\omega$ at rate $f$, a window $T_Z$ gives $\sigma_b = \sigma_\omega / \sqrt{f\,T_Z}$; the 10 s waypoint stops are such windows by rule.

**Slope.** The gyro measures rotation about the *body* $z$ axis; the heading needs rotation about the world vertical. With roll $\phi$ and pitch $\vartheta$ (ZYX Euler angles) and body rates $p, q, r$,

$$\dot{\psi} = \frac{q \sin\phi + r \cos\phi}{\cos\vartheta}. \tag{20}$$

Turning 90° on a ramp of 6° pitch while integrating $r$ alone turns in reality 90.50°; at 8° 90.88°. Over a following 40 m segment that is 35 cm and 62 cm of cross-track error — larger than the rest of the budget. Mitigations: (1) **turn only on level ground** (removes almost all of it); (2) project with Eq. (20), roll and pitch from a complementary filter on accelerometer and gyro (valid because specific force is dominated by gravity at these accelerations). Path cross-fall of 1 % to 2 % is irrelevant.

### 4.10 Range-sensor models (VL53L1X)

The VL53L1X is a 940 nm single-photon time-of-flight sensor with a 27° full field of view, a programmable region of interest (ROI, minimum 4 × 4 SPADs), distance modes short (≈ 1.3 m) to long (≈ 4 m indoors, much less in sunlight) and per-measurement outputs of range, *signal rate*, *ambient rate* and *sigma* (estimated range standard deviation). The measurement belongs to the middle of the integration window (timing budget), which matters for timestamping.

**Downward versus angled mounting.** Mounted at height $h$ and looking straight down, the sensor reads $r = h/(\cos\phi\cos\vartheta)$; pitch enters only to second order: at 5° the error is 0.38 %, 0.76 mm at $h = 200$ mm. Mounted at angle $\alpha$ to the ground it reads $r = h/\sin\alpha$ with first-order sensitivity

$$\frac{\partial r}{\partial \alpha} = -\frac{h\cos\alpha}{\sin^2\alpha} \;\Rightarrow\; 1.21\,\text{cm}/^\circ \text{ at } h = 20\,\text{cm},\ \alpha = 30^\circ, \tag{21}$$

so a 4° pitch on the ramp would move the reading by 4.8 cm — more than the 3 cm to 4 cm rubber edging it is meant to detect. Hence all ground sensors look straight down.

**Footprint.** The illuminated spot has diameter $2h\tan(\text{FoV}/2)$:

| | $h = 200$ mm | $h = 300$ mm |
|---|---|---|
| Full FoV 27° | 96 mm | 144 mm |
| ROI 4 × 4, FoV ≈ 15° | 53 mm | 79 mm |

A short baffle tube further limits stray light. The mounting height therefore directly sets the lateral resolution of the edge measurement; low mounting is better (100 mm to 200 mm) [DECIDE].

**Surface classification.** Grass is a diffuse, unevenly high scatterer: high sigma, low signal rate. Paving: low sigma, high signal rate. Rubber edging: low sigma at a different height. This two-feature, three-class discriminator is independent of colour and illumination and also works where there is no height step. [MEASURE] thresholds outdoors, noon and afternoon.

**Lateral heading fit.** A sideways-looking sensor passing a straight face (bench, kerb, wall) measures $r_i = d_0 - s_i \sin\Delta\psi + n_i$, with $\Delta\psi$ the heading relative to the face and $s_i$ the distance travelled. A least-squares line fit over a baseline $S$ with $N$ evenly spaced samples has slope variance $\sigma_r^2 / \sum (s_i - \bar{s})^2 \approx 12\,\sigma_r^2/(NS^2)$, so

$$\sigma_{\Delta\psi} \approx \frac{\sigma_r \sqrt{12}}{S\sqrt{N}}. \tag{22}$$

With pessimistic $\sigma_r = 2\,\text{cm}$, $S = 2\,\text{m}$, $N = 250$ (50 Hz, 0.4 m/s): $\sigma_{\Delta\psi} = 0.13^\circ$, i.e. 8.8 cm lateral over a following 40 m — against 21 cm for a 0.3° entry error without the fit. The absolute heading of the face must be known from the course survey to $\approx 0.5^\circ$ or better, or the fit only gives a relative heading.

**Cylindrical landmarks.** A post seen sideways produces a notch in the range profile; its centre (midpoint of the flanks) gives the along-track position. A wider spot widens the notch symmetrically but does not shift it, so the flank-midpoint method remains valid with the VL53L1X.

# 5 Measured parameters and identification

## 5.1 Bench session, week 5 (previous driver, no load)

Set-up: L298-style opto-isolated driver on a laboratory supply, one motor at a time, no wheel, bench firmware (`BENCH_README.md`) with a 1 kHz tick and raw-duty sweeps. Two sweeps per motor, nine duty steps from 20 % to 100 %, linear least-squares fit of duty on speed; breakaway from 3–4 slow ramps.

| Quantity | Motor L | Motor R | Note |
|---|---|---|---|
| $K_v$ [duty/rpm] | 0.00492 | 0.00507 | slope of the running fit |
| $K_s$ [duty] | 0.110 | 0.127 | intercept = kinetic friction |
| Breakaway duty | 0.141 | 0.169 | static friction, ±5 % scatter |
| No-load speed at $d = 1$ | 181 rpm | 177 rpm | catalogue 178 |
| Linearity | $R^2 > 0.999$ | $R^2 > 0.999$ | model Eq. (12) confirmed |
| Counts per output rev. | 3595 | 3618 | hand-turned, 3 rev. |
| Encoder A–B resistance | ≈6 kΩ | ≈6 kΩ | symmetric ⇒ 2 × 3 kΩ pull-ups via V+ |
| Encoder sign (bench wiring) | −1 | +1 | harness-dependent (vehicle: +1/ − 1) |

**Encoder V+ identification without power.** Because the open-collector pull-ups hang on V+, the wire showing ≈ 3 kΩ to *both* A and B is V+, an open wire to both is GND, and a pair reading a few ohms is the winding. Recording the 6 kΩ per motor at goods-in gives a later reference for a detached pull-up or a damaged output.

**Previous driver truth table** (6 V, 100 mA limit, motors disconnected): the state IN1 = IN2 = 1 drew no current, ruling out shoot-through; brake versus coast was not resolved. With the move to the BTS7960 this is closed: each BTS7960 is a half-bridge with internal dead time; RPWM = LPWM = 1 drives both outputs high (a brake to the supply rail), which is not destructive, but the firmware still never produces it.

## 5.2 Identification on the BTS7960 (9 Oct, wheels in the air)

Step and ramp experiments on both wheels, first-order-plus-dead-time fit:

| Quantity | Value | Note |
|---|---|---|
| Static gain $K$ | 7.596 (mm/s)/% | both wheels, slope of $v_{ss}(u)$ |
| Time constant $\tau_m$ | 50 ms | |
| Plant dead time $L$ | 0 ms | design delay $\theta = L + T_s +$ filter = 49 ms |
| Dead zone $u_0$ | 7.0 % | mean of both wheels |
| Feedforward L | 9.8 % + 0.1275 %/(mm/s) | per-wheel fit |
| Feedforward R | 4.1 % + 0.1361 %/(mm/s) | per-wheel fit |
| Highest speed | 680 mm/s at 100 % | |

**Plausibility of the per-wheel intercepts.** On the bench the right motor had the *higher* intercept (0.127 vs 0.110); on the BTS7960 it has the *lower* one (4.1 % vs 9.8 %), while the slope ordering (R weaker) is preserved. Slope and intercept of a line fit are negatively correlated, and a fit over a narrow speed range trades one against the other. [VERIFY] with the calibration sweep `K` (16 levels, 2 repetitions, standard errors reported) and the closed-loop ground sweep `G`; record $U_{bat}$ for both.

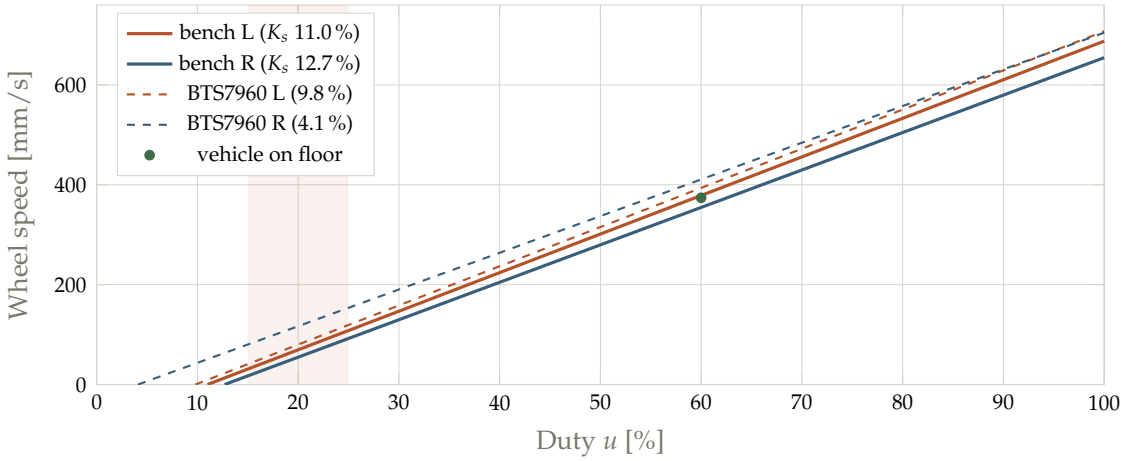



**Figure 5:** Duty-to-speed characteristics. Solid: bench (old driver, lab supply, no load), converted with $D = 72.6$ mm. Dashed: BTS7960 identification, wheels in the air. Dot: open-loop measurement on the floor at 60 % (8 Oct), 374 mm/s against 379 mm/s predicted by the bench model for the left wheel — the load on level paving costs about 1 %. The shaded band is the duty range of slow final approaches, where friction modelling matters most.

### 5.3 Vehicle tests (8 Oct)

| Test | L [mm] | R [mm] | Finding |
|---|---|---|---|
| `D1000`, TIM6 at 1 kHz (wrong) | 292.3 | 288.5 | tick 20× too fast ⇒ profile too short |
| `D1000`, PLLM /8, TIM6 41.7 Hz | 994.5 | 994.1 | settle timeout 5.5 mm before the target |
| `D1000` after clock fix, $D = 76$ mm | 999.5 | 1000.3 | firmware view; tape: 955 mm |
| ⇒ effective diameter | $76 \times 955/1000 = 72.59$ mm | | entered as 72.6 mm (with $C = 3584$) |
| `D2000`, `V800`, short ramps | 1766.3 | 1758.9 | saturation: 800 mm/s above the drive limit |
| Open loop 60 %, floor | 374 mm/s | — | bench model predicts 379 mm/s |

All firmware distances are encoder distances; only the 955 mm tape value is ground truth. Repeatability of the end position and lateral drift on long runs have not been measured [MEASURE].

### 5.4 Motor pairing

| Quantity (bench, equal duty) | L | R | Difference |
|---|---|---|---|
| Speed at 30 % duty | 39.3 rpm | 35.1 rpm | +12 % left |
| $K_v$ | 0.00492 | 0.00507 | 3 % |
| Breakaway duty | 0.141 | 0.169 | +20 % right |
| No-load speed | 181 rpm | 177 rpm | 2 % |
| Implied gear ratio | 56.17 | 56.53 | 0.64 % (see Section 4.3) |

The speed difference at low duty is dominated by friction (breakaway +20 %, slope only 3 %): at 30 % the left motor has more net voltage above its friction threshold. Consequences: open loop with a common duty the vehicle drives an arc; the per-wheel feedforward removes most of it before any loop closes; the per-wheel speed loop removes the rest; matching motors by purchase would be expensive and would drift with wear. [VERIFY] Re-measure with wheels, vehicle mass and battery; load friction may not scale equally on both sides.

## 5.5 Calibration status

| Constant | Status | Source / next step |
|---|---|---|
| Encoder signs, motor signs | done | vehicle: `ENC_L/R = +1/-1`, `MOT_L/R = -1/+1`; re-check with `C` after any harness change |
| $K_v$, $K_s$ (no load) | done | bench + BTS7960 wheels-in-air |
| $K_v$, $K_s$ under load | open | calibration `G` on the floor |
| Breakaway on BTS7960 | open | `B`; code still uses bench 14.1/16.9 % |
| $\tau_m$ under load | open | `G/K` level changes |
| Mean scale $D/C$ | done | 72.6 mm / 3584; repeat `D5000`×3 per surface |
| Per-wheel mm/count ($E_d$) | open | UMBmark `Q±2000`×5 |
| Effective wheelbase ($E_b$) | open | UMBmark, `W±3600` |
| Backlash $\delta_b$ | open | `L` |
| Driver duty floor $U_{min}$ | open | oscilloscope on the BTS7960 output |
| Battery divider ratio | open | multimeter vs ADC |
| Gyro bias, ARW, temperature | after IMU | Allan variance, temperature run |
| VL53L1X thresholds, $\sigma_r$ | after fitting | outdoor afternoon test |

# 6 Control system

## 6.1 Structure of the wheel cascade (implemented)

Each wheel has its own cascade, executed in the TIM6 interrupt at 50 Hz (`Control_Tick()`): a time-based trapezoidal profile generates $v_{prof}(t)$ and $x_{prof}(t)$; a P position loop corrects deviations from $x_{prof}$; an inner speed loop with model feedforward and PI feedback produces the duty. The two cascades are coupled by a synchronisation term that keeps the wheel position errors equal, i.e. suppresses heading transients.

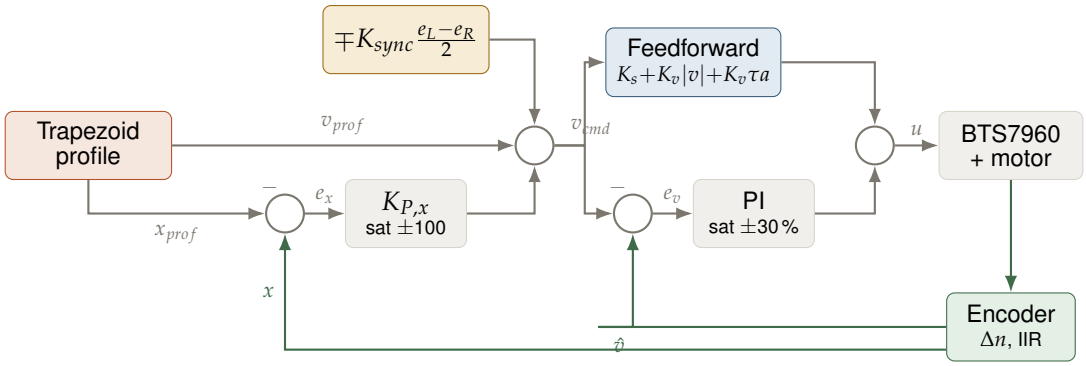



**Figure 6:** Wheel cascade per side (L and R identical, own constants). The feedforward carries the bulk of the duty; the PI only corrects model error; the position loop only corrects the integral of what the speed loop missed.

## 6.2 Actuation layer

- **Mapping.** $d \cdot \sigma_{mot} > 0$ ($\sigma_{mot}$ = `MOT_x_SIGN`) ⇒ RPWM carries the PWM and LPWM is 0; otherwise the reverse. Both enables of a module are bridged on one GPIO (EN_L = PB12, EN_R = PB13). A single setter function is the only writer of CCR1–4 and the EN lines; it sets the inactive channel to 0 *before* the active one, so no instruction-length state with both inputs high exists.
- **Frequency.** 20 kHz — inaudible and within the BTS7960's 25 kHz limit. The 5-vs-10 kHz decision of the concept (old driver with a 10 kHz limit and a large duty floor) is superseded. [MEASURE] the real output floor $U_{min}$ at 20 kHz with an oscilloscope; if it approaches the 7 % dead zone, 10 kHz halves it at the cost of audible noise.
- **Reversal.** The bench firmware enforces 50 ms coast on every reversal and a slew limit of 0→100 % in 500 ms; the vehicle firmware avoids end-of-move reversals altogether (backlash, Section 4.3).
- **Defined off-state.** EN lines have internal and external (10 kΩ) pull-downs; TIM1 idle state is "reset" (outputs low while MOE is cleared). `Error_Handler` must actively clear CCR1–4 and pull EN low (open item O-12) — otherwise a fault leaves PWM running with enables high.

## 6.3 Wheel speed control

**Control law.** For each wheel, with $v_{cmd}$ the commanded speed, $\hat{v}$ the filtered measured speed and $e_v = v_{cmd} - \hat{v}$:

$$u = \underbrace{\operatorname{sgn}(v_{cmd})\left(K_s^* + K_v|v_{cmd}|\right) + K_v\,\tau_m\,a_{prof}}_{\text{feedforward}} + \underbrace{K_p\,e_v + I_k}_{\text{PI}},$$
$$I_k = \operatorname{sat}_{\pm 30\,\%}\left(I_{k-1} + K_i\,e_v\,T_s\right). \tag{23}$$

- $K_s^* = K_{s,break}$ while the wheel stands ($|\hat{v}| < 3\,\mathrm{mm/s}$), $K_s$ (running intercept) once it rolls ($|\hat{v}| > 8\,\mathrm{mm/s}$), with hysteresis in between. This is the friction model of Section 4.5 made explicit; it replaced the earlier "kick" (40 % for up to 5 ticks), which acted per wheel independently and let one wheel break away before the other — a heading error at every start.
- $K_v \tau_m a$ is the inertia feedforward: for the first-order plant Eq. (13), the duty that produces $\dot{v} = a$ on top of steady state is $\tau_m a / K$, and $1/K = K_v$.
- Unit conversion of the bench constants: $K_v^{[\%/(mm/s)]} = K_v^{[duty/rpm]} \cdot 100 \cdot 60/(\pi D)$, i.e. 0.1294 (L) and 0.1334 (R) %/(mm/s).

**Tuning by SIMC.** For a first-order plant with dead time, $G(s) = Ke^{-\theta s}/(\tau s + 1)$, the SIMC rules (Skogestad) give

$$K_p = \frac{\tau}{K(\tau_c + \theta)}, \qquad T_i = \min\big(\tau,\, 4(\tau_c + \theta)\big), \qquad K_i = K_p / T_i, \tag{24}$$

with the closed-loop time constant $\tau_c$ as the single tuning knob. With $K = 7.596\,(\mathrm{mm/s})/\%$, $\tau = 50\,\mathrm{ms}$ and design delay $\theta = 49\,\mathrm{ms}$ (sample period plus the IIR filter):

| **Setting** | $\tau_c$ | $K_p$ [%/(mm/s)] | $K_i$ [%/(mm/s s)] | $T_i$ | overshoot | settling 5 % |
|---|---|---|---|---|---|---|
| tight | $\theta$ | 0.067 | 1.34 | 50 ms | 4.2 % | 200 ms |
| **balanced** | $2\theta$ | **0.045** | **0.90** | 50 ms | 0.3 % | 320 ms |
| gentle | $4\theta$ | 0.027 | 0.54 | 50 ms | 0.0 % | 640 ms |
| code state 9 Oct | — | 0.10 | 0.50 | 200 ms | not designed | |

With $T_i = \tau$ the PI zero cancels the plant pole and the loop transfer function becomes $L(s) = e^{-\theta s}/\big((\tau_c + \theta)s\big)$. For the balanced setting: crossover $\omega_c = 1/(\tau_c + \theta) = 6.8\,\mathrm{rad/s}$ (1.1 Hz), phase margin $90^\circ - \omega_c\theta \cdot \frac{180}{\pi} = 71^\circ$, gain margin $\pi/(2\theta\,\omega_c) = 4.7$ (13.4 dB). Simulated with the feedforward on a 1 s ramp to 200 mm/s: maximum speed error 13.1 mm/s for the plant as identified, 7.7 mm/s at −20 % gain and 17.1 mm/s at +20 %. The revised firmware (10 Oct) uses the balanced setting. [VERIFY] which build is flashed; the spreadsheet of 9 Oct still lists 0.10/0.50.

**Anti-windup.** The integrator is clamped to ±30 % duty (`INTEG_MAX_PCT`). Because the feedforward carries the operating point, the integrator only needs to cover model error and load; 30 % also bounds the damage of a stalled wheel.

### 6.4 Trajectory generation and wheel position control

**Trapezoidal profile.** For a distance $D$, nominal speed $V$, acceleration time $t_a$ and deceleration time $t_d$, the move time is

$$T = \frac{D}{V} + \frac{t_a + t_d}{2}, \qquad \text{or, if } D < \tfrac{V}{2}(t_a + t_d): \quad V' = \frac{2D}{t_a + t_d},\ T = t_a + t_d \text{ (triangle)}. \tag{25}$$

`Profile_Eval()` returns $v_{prof}(t)$ and $x_{prof}(t) = \int v_{prof}\,\mathrm{d}t$ at every tick. Defaults: $V = 300\,\mathrm{mm/s}$, $t_a = t_d = 1000\,\mathrm{ms}$ ($a = 0.3\,\mathrm{m/s^2}$); range 5 mm to 20 000 mm.

**Position loop.**

$$e_x = \mathrm{dir} \cdot x_{prof} - x, \qquad v_{cmd} = \mathrm{dir} \cdot v_{prof} + \mathrm{sat}_{\pm 100\,mm/s}\big(K_{P,x}\, e_x\big), \qquad K_{P,x} = 2\,\mathrm{s^{-1}}. \tag{26}$$

With an ideal inner loop, $\dot{e}_x = -K_{P,x} e_x$, so position errors decay with $\tau = 0.5\,\mathrm{s}$. No integral term is needed: the wheel integrates speed to position, and with speed feedforward a ramp leaves no following error. The bandwidth ratio to the speed loop ($\omega_c = 6.8\,\mathrm{rad/s}$ vs $2\,\mathrm{rad/s}$) is 3.4 — slightly below the rule-of-thumb 5; acceptable because the feedforward carries the motion [VERIFY] by overshoot at the end of `D300`.

**End of move.** `SETTLE` when the profile has finished: done when $|e_x| < 2\,\mathrm{mm}$ and $|\hat{v}| < 20\,\mathrm{mm/s}$; after 50 ticks (1 s) the move ends regardless. Each wheel is braked on arrival; the controller never reverses at the end (backlash). Two known weaknesses: the timeout currently also reports `DONE` (O-6), and with $K_s$ below the breakaway value a wheel can stop a few millimetres short (O-2) — the reason for the breakaway switching above.

### 6.5 Left/right synchronisation

The heading of the vehicle is $\theta = (s_R - s_L)/b$ at every instant, not only at the end. Two independent cascades with equal final positions can still leave a lateral offset if one wheel lags during acceleration. The revised firmware couples them:

$$v_{cmd,L} \mathrel{+}= \tfrac{1}{2} K_{sync}(e_{x,L} - e_{x,R}), \qquad v_{cmd,R} \mathrel{-}= \tfrac{1}{2} K_{sync}(e_{x,L} - e_{x,R}). \tag{27}$$

With $\Delta = e_{x,L} - e_{x,R}$ and ideal inner loops, $\dot{\Delta} = -(K_{P,x} + K_{sync})\Delta$: the *differential* mode (heading) decays faster than the common mode (distance). With $K_{sync} = 3\,(\mathrm{mm/s})/\mathrm{mm}$ the heading-error time constant is 0.2 s against 0.5 s for distance. Sizing criterion: the maximum L/R tracking difference on `D1000/D5000` should stay below 3 mm [MEASURE]. Once the EKF exists the heading is closed by the path follower on the gyro heading; synchronisation then remains as the fast inner differential loop.

### 6.6 Path following

**Path, not trajectory.** The reference is a geometric path with a speed profile, decoupled from time. An obstacle that delays the vehicle by 40 s does not create a "catch-up" error.

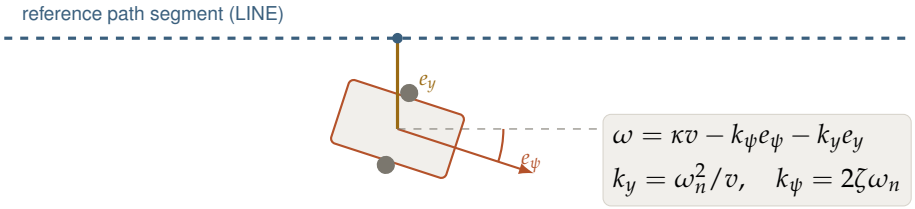



**Figure 7:** Cross-track error $e_y$ (signed, left of the path positive) and heading error $e_\psi = \theta - \theta_{path}$. The feedforward $\kappa v$ follows the path curvature; the feedback is a linear second-order design.

**Linear cross-track/heading controller (recommended).** Relative to a path of curvature $\kappa$ the error kinematics are $\dot{e}_y = v \sin e_\psi$, $\dot{e}_\psi = \omega - \kappa v$. With $\omega = \kappa v - k_\psi e_\psi - k_y e_y$ and small angles,

$$\ddot{e}_y + k_\psi \dot{e}_y + k_y v\, e_y = 0 \quad \Rightarrow \quad \omega_n^2 = k_y v, \qquad 2\zeta\omega_n = k_\psi. \tag{28}$$

Scheduling $k_y = \omega_n^2 / v$ keeps the dynamics independent of speed (in distance terms the convergence then shortens at low speed — acceptable, because slow driving only happens in approaches). **Bandwidth choice:** the speed loop closes at $\omega_c = 6.8\,\mathrm{rad/s}$ and the wheel position loop at $2\,\mathrm{rad/s}$; with a factor of 3–5 separation the path loop should use $\omega_n \approx 1.5\,\mathrm{rad/s}$ to $2\,\mathrm{rad/s}$, $\zeta = 0.7$, i.e. $k_\psi = 2.1\,\mathrm{s}^{-1}$ to $2.8\,\mathrm{s}^{-1}$ and $k_y = 7.5\,\mathrm{m}^{-1}\mathrm{s}^{-1}$ to $13.3\,\mathrm{m}^{-1}\mathrm{s}^{-1}$ at 0.3 m/s. (The earlier concept figure

of "about 1 Hz" bandwidth, 6.3 rad/s, would sit on top of the speed loop and is corrected here.) The command is limited so that neither wheel set-point $v \pm \omega b/2$ exceeds the speed ceiling; if it would, $v$ is reduced and $\omega$ kept.

**Pure Pursuit (alternative).** Steer towards a look-ahead point at distance $L_d$: $\kappa = 2y_L/L_d^2$ ($y_L$ lateral offset of that point in the body frame), $\omega = \kappa v$, $L_d = L_0 + kv$. Robust with one parameter, but it cuts corners systematically; kept as fallback.

**TURN primitive.** Corners are driven as stop–turn–go. The turn follows a trapezoidal angle profile $\theta_{ref}(t)$ with $\omega_{cmd} = \dot{\theta}_{ref} + K_\theta(\theta_{ref} - \hat{\theta})$ on the gyro heading, wheel set-points $\pm\omega_{cmd} b/2$, $K_\theta \approx 2\,\mathrm{s}^{-1}$; termination on $|\theta_{ref,end} - \hat{\theta}| < 0.2^\circ$ and $|\omega|$ small, approaching from one direction only. Turn in place is the cleanest manoeuvre of a differential drive: ICR at the axle midpoint, no lateral tyre forces, best gyro accuracy (pure yaw).

**Mapping onto the present firmware.** A `LINE` of known length maps onto today's `D<mm>` cascade with the path follower's $\omega$ added as a differential speed $\pm\omega b/2$ to $v_{cmd,L/R}$; a `TURN` maps onto a distance move with opposite signs, $D_{L,R} = \mp\Delta\theta\, b/2$, corrected by the gyro. This reuses the tested wheel cascade instead of replacing it.

## 6.7 Mission logic

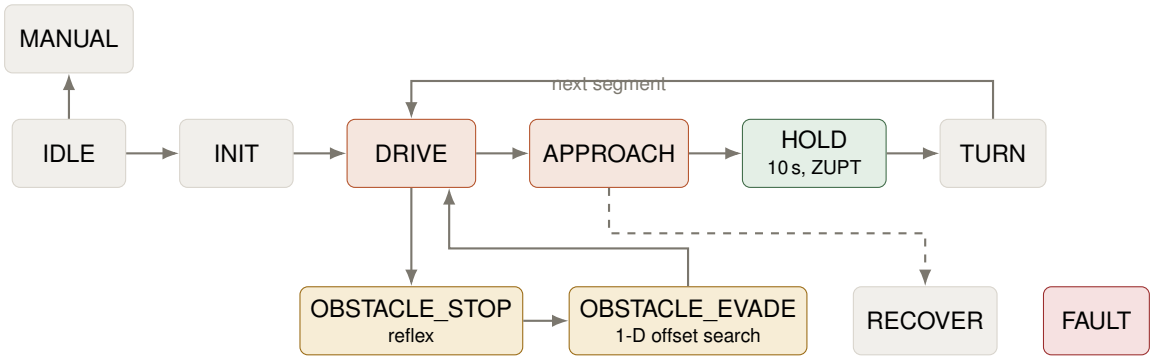



**Figure 8:** Mission state machine. Each state has entry, cycle and exit actions and an explicit table of allowed transitions; no nested if-cascades.

| Transition | Guard and rationale |
|---|---|
| INIT → DRIVE | gyro bias variance $P_{b_g}$ below threshold (not a fixed timeout); battery above cut-off; all sensors alive |
| DRIVE → APPROACH | along-track distance to the waypoint $< s_{appr}$ (1 m) **and** $\lvert y_{offset}\rvert < 0.05$ m: an unfinished evasion keeps the vehicle in DRIVE — seconds are free, centimetres are graded |
| APPROACH → HOLD | $\lvert e_s\rvert < 2$ mm in the encoder frame and speed zero; approach at 0.1 m/s, always ending forward |
| HOLD → TURN | 10 s elapsed (timed from standstill, with margin 0.5 s); ZUPT applied; bias updated |
| any → OBSTACLE_STOP | forward range below the stopping distance — issued by the reflex layer, not by this machine |
| OBSTACLE_STOP → EVADE | obstacle persists beyond a wait time (temporary obstacles are waited out: 1 point; rerouting: 2 points) |
| any → RECOVER | reference lost too long, or tr $P$ above a bound; slow down, seek the edge |
| any → FAULT | stall, battery cut-off, sensor timeout, inner ground sensor sees drop/grass |

### 6.8 Safety and failsafe

| Mechanism | Behaviour | State |
|---|---|---|
| EN pull-downs | internal + 10 kΩ external: drivers off during reset and before init | done |
| Single setter | forbidden output combinations unreachable by construction | done |
| Stall detection | bench: $\lvert d\rvert > 15$ %, $< 5$ rpm, 300 ms; calibration: $\lvert d\rvert \geq 0.30$, $< 0.5$ rpm, 0.5 s; **suspended** during deliberate-stall measurements | done |
| Wheel fault | position error persists 0.6 s (30 ticks) → stop | done |
| Dead-man | no input for 2 s in open-loop/manual → stop | done |
| Tracking/excursion abort | calibration: tracking error $> 40$ mm, L/R difference $> 30$ mm | done |
| Radio heartbeat | manual mode: no valid command for 500 ms → $v_{ref} = 0$ | planned |
| Battery cut-off | 10.0 V, plus independent LiPo alarm on the balance lead | divider missing |
| Error_Handler | CCR1–4 = 0, EN low before looping | O-12 |
| IWDG | LSI 32 kHz, prescaler 32, reload 100 ⇒ 100 ms; frozen in debug (`__HAL_DBGMCU_FREEZE_IWDG`) | planned |
| CSS | clock failure → HSI + NMI; NMI handler stops the motors | planned |
| Reflex stop | forward VL53L1X, highest priority, bypasses planning | planned |
| Inner ground interlock | inner downward sensor sees drop/grass → stop; never disabled, not even during evasion | planned |
| Hardware kill | **possible now**: EN lines are separate from PWM on the BTS7960, so a mechanical E-stop in series with EN (or the main switch) gives a firmware-independent stop [DECIDE] | proposal |

**Lesson from the bench.** The stall supervisor aborted the deadband sweep, because a stationary shaft above the duty threshold *is* the measurement there. A safety mechanism and a measurement procedure contradicted each other. Every new sequence that deliberately stalls a wheel (breakaway test, pushing against a kerb on purpose) must re-examine the stall rule's scope.

# 7 State estimation

The controller can hold a position to a few millimetres in its *own* frame; the graded quantity is where the vehicle really is. The accuracy grade therefore rides on the estimate, not on the controller. The estimator is a five-state Extended Kalman Filter running at 200 Hz [OPEN].

## 7.1 State and process model

$$\boldsymbol{x} = \begin{bmatrix} x & y & \theta & b_g & s \end{bmatrix}^\top \tag{29}$$

| State | Meaning | Why it is estimated |
|---|---|---|
| $x$, $y$ | position of the axle midpoint in $\{W\}$ | the graded quantity |
| $\theta$ | heading | drives all position integration |
| $b_g$ | gyro bias | drifts with temperature; integrates directly into heading error |
| $s$ | odometry scale factor ($\approx 1$) | largest along-track error; changes with load, tyre and surface |

**Inputs** per step $\Delta t$: the encoder distance $\Delta s_e = \frac{1}{2}(\Delta s_L + \Delta s_R)$ with nominal distance per count, and the mean gyro rate $\omega_m$ over the step. Heading comes from the gyro; the wheel difference is used only by the slip detector.

$$\begin{aligned} \varphi &= \theta + \tfrac{1}{2}(\omega_m - b_g)\Delta t, \quad \Delta s = s\,\Delta s_e, \\ x^+ &= x + \Delta s \cos\varphi, \qquad\quad y^+ = y + \Delta s \sin\varphi, \\ \theta^+ &= \theta + (\omega_m - b_g)\Delta t, \quad b_g^+ = b_g, \quad s^+ = s. \end{aligned} \tag{30}$$

Bias and scale are random walks. Jacobians with respect to the state and to the inputs $(\Delta s_e, \omega_m)$, using $c_\varphi = \cos\varphi$, $s_\varphi = \sin\varphi$:

$$F = \begin{bmatrix} 1 & 0 & -\Delta s\, s_\varphi & \tfrac{1}{2}\Delta s\, s_\varphi \Delta t & \Delta s_e c_\varphi \\ 0 & 1 & \Delta s\, c_\varphi & -\tfrac{1}{2}\Delta s\, c_\varphi \Delta t & \Delta s_e s_\varphi \\ 0 & 0 & 1 & -\Delta t & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} s\, c_\varphi & -\tfrac{1}{2}\Delta s\, s_\varphi \Delta t \\ s\, s_\varphi & \tfrac{1}{2}\Delta s\, c_\varphi \Delta t \\ 0 & \Delta t \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{31}$$

$$P^+ = FPF^\top + GMG^\top + Q_{rw}, \qquad M = \operatorname{diag}\!\left(k_d|\Delta s_e|,\ \sigma_\omega^2\right), \qquad Q_{rw} = \operatorname{diag}\!\left(0,0,0,\ q_b\Delta t,\ q_s\Delta t\right). \tag{32}$$

The distance noise grows with distance travelled ($k_d$, from repeated straight runs); $\sigma_\omega$, $q_b$ come from the IMU's Allan variance (angle random walk and bias instability); $q_s$ is small (the scale changes with surface, not with time).

**Slip detection.** The encoder yaw rate $\omega_e = (\Delta s_R - \Delta s_L)/(b\,\Delta t)$ is compared with the gyro, $r_{slip} = \omega_e - (\omega_m - b_g)$. If $|r_{slip}|$ exceeds a threshold for several samples (wheel spin, a caster dropping off an edge), $k_d$ is inflated for those samples: the filter trusts the encoders less exactly when they lie. It costs nothing and uses sensors that are already there.

## 7.2 Measurement models

All updates are scalar or 2-D and are processed sequentially, so no matrix inversion is needed. Sensor positions on the body are $\boldsymbol{r}_s$ (mounting point) and $\boldsymbol{u}_s$ (unit viewing direction); $R(\theta)$ is the 2-D rotation and $R'(\theta) = \mathrm{d}R/\mathrm{d}\theta$.

| Type | Model $h(\boldsymbol{x})$ | $H$ (non-zero) | When / noise |
|---|---|---|---|
| ZUPT | $h = b_g$, $z = \overline{\omega_m}$ over the stop | $H_{b_g} = 1$ | every stop $\geq 1$ s; $R = \sigma_\omega^2 / N$ |
| Edge crossing | $h = \boldsymbol{n}^\top(\boldsymbol{p} + R(\theta)\boldsymbol{r}_s) - d_e$, $z = 0$ | $\boldsymbol{n}^\top$; $\boldsymbol{n}^\top R' \boldsymbol{r}_s$ | downward sensor changes class (path/edge/grass); $\sigma \approx 2$ cm (spot) |
| Range to a face | $h = \dfrac{d_f - \boldsymbol{n}^\top(\boldsymbol{p} + R\boldsymbol{r}_s)}{\boldsymbol{n}^\top R\, \boldsymbol{u}_s}$ | $-\boldsymbol{n}^\top / g$; $\partial h / \partial \theta$ | lateral sensor along a mapped straight face; $g = \boldsymbol{n}^\top R \boldsymbol{u}_s$ |
| Face heading | $h = \theta$, $z = \psi_{face} + \Delta\hat{\psi}$ | $H_\theta = 1$ | after a line fit over $\geq 1$ m; $\sigma$ from Eq. (22) |
| Landmark (post) | $h = \boldsymbol{p} + R(\theta)(\boldsymbol{r}_s + r\,\boldsymbol{u}_s)$, $z = \boldsymbol{p}_L$ | $I_2$; $R'(\boldsymbol{r}_s + r\boldsymbol{u}_s)$ | notch centre in the lateral range profile; 2-D, makes $s$ observable |

**About the non-holonomic constraint.** The concept lists NHC (zero lateral body velocity, Eq. (4)) as a measurement type. In this five-state model the prediction Eq. (30) moves the vehicle only along its heading, so NHC is *already built into the process model* and adds no information as a separate update. It becomes a genuine pseudo-measurement only if the state is extended by velocity or a lateral slip state (e.g. for accelerometer-aided prediction). The listing is kept for completeness; no update code is needed for it now.

**Update.** $\nu = z - h(\hat{\boldsymbol{x}})$, $S = HPH^\top + R$, $K = PH^\top S^{-1}$, $\hat{\boldsymbol{x}} \mathrel{+}= K\nu$, and the covariance in Joseph form

$$P = (I - KH)P(I - KH)^\top + KRK^\top, \tag{33}$$

because the short form $(I - KH)P$ loses symmetry and positive definiteness in single precision — the F446 has a single-precision FPU only.

**Outlier gate.** Every update is gated on the Mahalanobis distance $d^2 = \nu^\top S^{-1}\nu$: reject if $d^2 > 9$ for a scalar ($\chi^2_1$, 99.7 %) or $d^2 > 9.2$ for 2-D ($\chi^2_2$, 99 %). A false edge detection or another vehicle standing where a landmark is expected would otherwise wrench the filter. **The gate is present from the first line of code.** Landmark and face updates are additionally suppressed while an evasion offset $|y_{offset}| > 0.1$ m is active.

**Timing.** Every measurement carries the timestamp of its physical acquisition from the 1 MHz time base: IMU by input capture of DRDY, VL53L1X at the middle of the integration window. Measurements older than the current state are applied with the state from a short history buffer (100 ms), or — for latencies below one EKF step — simply at the next step. Load: a five-state EKF at 200 Hz costs well under 1 % of the CPU.

**Initialisation.** INIT stands still until the variance of $b_g$ has converged; $x, y, \theta$ are zero by definition (start mark and start heading define $\{W\}$); $s = 1$ with the variance of the scale calibration. The start heading must therefore be set physically: a jig or two marks on the ground at the start position [DECIDE].

### 7.3 Observability

| Configuration | Observable | Not observable |
|---|---|---|
| Dead reckoning only | nothing absolute | $x, y, \theta$ drift; $s$ never |
| + ZUPT | $b_g$ | position, heading, $s$ |
| + lateral edge | position *perpendicular* to the edge | position along the edge |
| + straight-face fit | heading, absolutely | — |
| + longitudinal landmark | position along travel, hence $s$ | — |

**Flat surfaces give heading; round objects give position.** A post looks the same from every direction: a full 2-D position fix, no heading. A bench face gives heading to a fraction of a degree, but only a weak position fix. **Corners rotate the observable direction:** on a north-going segment the edge constrains $x$; after a 90° turn the same kind of edge constrains $y$ — precisely the error that accumulated unobserved before. Over the circuit the full 2-D position becomes observable without any absolute position sensor; the course geometry does it.

### 7.4 IMU integration

- **Selection.** ICM-42688-P unobtainable; **ISM330DHCX** (Adafruit breakout) selected. Criteria in order: gyro temperature-offset stability, noise density, SPI for DMA, a data-ready output for hardware timestamping, local availability. MPU-6050/9250 and ICM-20948 only as bring-up placeholders (poor temperature stability, discontinued, counterfeit-prone).
- **Interface.** SPI1 (PB3/PB4/PB5, CS PA4), 5.625 MHz (APB2/16; the ISM330DHCX allows at most 10 MHz); DRDY on PA1 = TIM2_CH2 input capture gives a jitter-free timestamp; the capture interrupt starts the DMA read. The ISM330DHCX has no 1 kHz output data rate: sample at 833 Hz with the sensor's anti-aliasing filter and decimate by 4 to $\approx$ 208 Hz for the 200 Hz EKF [VERIFY] (datasheet ODR table).
- **Mounting.** Rigid, at the axle midpoint (no lever-arm accelerations in turns), not soft-mounted (an elastic mount creates a resonance in the band of interest), thermally away from the drivers, under a light-coloured cover.
- **Temperature.** Outdoors the die temperature may move 20 K to 30 K during a run. Second-order polynomial bias model against the internal temperature sensor from a calibration run; 10 min warm-up before runs; ZUPT at every stop removes the residual.

# 8 Lateral aiding and boundary sensing

Lateral aiding (cross-track support) is the use of the path's own boundaries — grass edge, rubber edging, kerb, step, benches — as a continuous reference that bounds the cross-track error independently of how far the vehicle has driven. Dead reckoning drifts without bound; a boundary measurement does not.

## 8.1 Sensor layout and roles

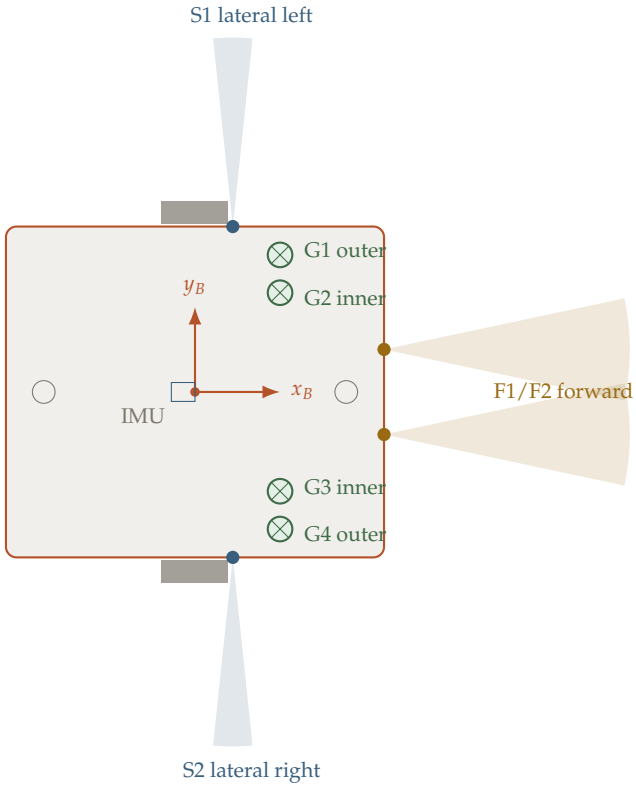



**Figure 9:** Planned sensor layout (top view, not to scale). Ground sensors G1–G4 look straight down, two per side at different lateral offsets (e.g. 15 and 25 cm from the centre line) [DECIDE]. Forward sensors F1/F2 serve the reflex layer and free-space estimation; lateral sensors S1/S2 look sideways at benches, posts and kerbs. G on I2C1, F and S on I2C2; every sensor has its own XSHUT line. Only five VL53L1X were ordered (four fitted, one spare): G1–G4 are fitted, F1/F2 and S1/S2 are not covered [DECIDE].

The four ground sensors have four distinct roles that are easy to conflate:

| # | Role | What it protects |
|---|---|---|
| 1 | Run validity | a circuit that touches grass is invalid — lost as a whole, not gradually |
| 2 | Physical safety | kerbs and the step are drop-offs; driving off ends the run and may damage the vehicle |
| 3 | EKF measurement | edge updates make the position perpendicular to the edge observable (Section 7.3) |
| 4 | Evasion bound | the lateral search range ends at the *detected* edge, not the estimated one (Section 9.6) |

No horizontal sensor can take these over: a 3 cm to 4 cm rubber edging, an 8 cm kerb or a flush grass transition is invisible to a sensor mounted 25 cm to 35 cm high and looking horizontally.

### 8.2 Three-valued edge state

| Inner sensor | Outer sensor | Meaning |
|---|---|---|
| path | path | edge further out than the outer sensor — drift away is allowed up to the corrido |
| path | edge/drop/grass | edge between the two sensors — **nominal**, cross-track known to $\approx \pm 5$ cm |
| edge/drop/grass | any | edge too close — **emergency stop** |

During an evasion the vehicle deliberately approaches an edge: the **outer** sensor is then the operating signal that bounds the offset; the **inner** sensor is the interlock and is **never disabled**. A planner that commands an offset which trips the inner sensor has a fault, not a manoeuvre. This is why both pairs are needed from the first autonomous build.

### 8.3 Classification of path, edge and grass

Each reading delivers range $r$, signal rate $S$ and sigma $\sigma$. With a mounting height $h_0$ over paving:

- **Drop/step down**: $r > h_0 + \Delta_{min}$ (kerb 8 cm, step);
- **Edging/step up**: $r < h_0 - \Delta_{min}$ with low $\sigma$;
- **Grass**: $r$ noisy and short (blades above ground), high $\sigma$, low $S$;
- **Paving**: $r \approx h_0$, low $\sigma$, high $S$.

A debounce of 2–3 consecutive samples suppresses single leaves or cracks; a transition is time-stamped at the middle of the integration window of the first sample in the new class. [MEASURE] thresholds $\Delta_{min}$, $\sigma$ and $S$ on the actual surfaces at noon and in the afternoon (sunlight raises the ambient rate and $\sigma$).

### 8.4 Cross-track correction

Two forms are used, depending on the boundary:

1. **Edge crossing events** (ground sensors). Each class change of a sensor places its footprint on the mapped boundary line; the EKF receives the constraint $\boldsymbol{n}^\top(\boldsymbol{p} + R(\theta)\boldsymbol{r}_s) = d_e$ (Section 7.2). With the vehicle driving along the edge, the outer sensor toggles intermittently as it skims the edge, which gives a stream of fixes. Accuracy is set by the footprint (5 cm to 8 cm at ROI $4 \times 4$, Section 4.10), $\sigma \approx 2$ cm.
2. **Continuous range** (lateral sensors) to a face of known position: kerb face, bench face, wall. Gives cross-track continuously and, via the line fit (Eq. (22)), absolute heading.

> **Why the boundary matters more than any other aid.** The entry heading error at the start of a reference-free segment dominates the error budget (Section 12); a boundary bounds cross-track drift wherever it exists, and a straight face resets heading. The path is in effect a rail: the edge is a free measurement that is always present unless the vehicle is in an evasion or the boundary has a gap. Reference-free stretches immediately before waypoints are therefore the decisive survey item [MEASURE].

### 8.5 Landmarks for along-track position

Along a straight edge, position *along* the edge is unobservable. It is supplied by discrete landmarks: posts (2-D fix via the notch centre), the ends of benches and tables, the beginning or end of

an edging, path junctions (edge appears or disappears on one side), and the step (also detectable as an IMU impulse if crossed). The landmark map is a survey product: positions to $\pm 5$ cm, face orientations to $\pm 0.5^\circ$ [MEASURE]. Furniture may be moved before the exam day [ASK]; landmark updates are gated and optional by design.

### 8.6 Bus and timing

- Four sensors per bus at 400 kHz; a timing budget of 20 ms gives 50 Hz per sensor; with the Ultra Lite Driver each readout is a few dozen bytes, bus load $\approx 15\,\%$.
- Start-up: all XSHUT low, then release one at a time and assign a new address.
- One hung device blocks a bus: per-transaction timeouts and a bus recovery (nine clock pulses, STOP). Ground and horizontal sensors on separate buses so that one failure does not take out edge *and* obstacle detection.
- The two forward sensors start with a time offset so that their emitters do not disturb each other where the fields of view overlap.
- [RISK] Outdoor range: short mode is nominally 1.3 m, at noon in sun less. For the reflex stop 0.5 m suffice; for evasion planning the vehicle slows to 0.25 m/s when an object is in view.

# 9 Path planning

## 9.1 Course representation

The course is known in advance; planning is therefore an *offline* geometric task done once from the survey, plus an *online* 1-D lateral re-planning for obstacles. The path is a list of primitives in $\{W\}$:

$$\{\ \texttt{LINE}(\boldsymbol{p}_1, \boldsymbol{p}_2)\ |\ \texttt{ARC}(\boldsymbol{c}, r, \varphi_1, \varphi_2)\ |\ \texttt{TURN}(\theta_{target})\ |\ \texttt{PAUSE}(t)\ |\ \texttt{WAYPOINT}(id)\ \}$$

Each segment also carries its **corridor** (signed distances from the path line to the left and right boundaries, boundary type per side), its **speed limit**, and the **landmarks** visible from it. A plain polyline through the waypoints is not used: its curvature jumps at every vertex would demand infinite $\omega$.

## 9.2 Route

From the schematic (Fig. 1), clockwise around the central green:

| Leg | From → to | Character | Length |
|---|---|---|---|
| 1 | X → WP1 (south) | diagonal path south-west, then west into the waypoint | [MEASURE] |
| 2 | WP1 → WP2 (north-west) | long straight northbound path on the western side — probably the longest reference segment | [MEASURE] |
| 3 | WP2 → corner north-east | eastbound along the northern path, step running parallel on the north side | [MEASURE] |
| 4 | corner → WP3 (north-east) | short south-east leg | [MEASURE] |
| 5 | WP3 → X | south, then diagonal south-west back to the start; the barrier lies south-east of X | [MEASURE] |

Total length ≈ 150 m (estimate from the concept; [MEASURE]). Corners between legs are driven as stop–`TURN`–go on level ground; `ARC` only where the path itself is curved.

## 9.3 Speed profile and time budget

| Cruise speed | 0.20 | 0.25 | 0.30 | 0.40 | m/s |
|---|---|---|---|---|---|
| Driving time for 150 m | 750 | 600 | 500 | 375 | s |
| + 3 waypoint stops (10 s) | 780 | 630 | 530 | 405 | s |
| Reserve to 15 min (turns, approaches, obstacles) | 120 | 270 | 370 | 495 | s |

Recommended: cruise 0.3 m/s, 0.25 m/s when an object is in the forward field, 0.1 m/s over the last metre to a waypoint, accelerations $\leq 0.3\,\mathrm{m/s^2}$. This leaves about 6 minutes for turns, slow approaches, INIT and waiting out obstacles. Driving faster earns nothing and costs accuracy (scale error grows with slip; transients grow with acceleration).

## 9.4 Waypoint approach and hold

Waypoint accuracy is made in the last metre, not by the path follower:

1. Before the waypoint, use the last reliable references: a landmark fix for along-track, the edge or a face fit for cross-track and heading.

2. `APPROACH` at 0.1 m/s with the Coulomb feedforward active (the weaker wheel breaks away near the approach duty), deceleration ending forward, no reversal.
3. Stop on the EKF estimate of the waypoint position; hold 10 s plus a margin, apply ZUPT (bias reset for the next leg).
4. If an obstacle stands on the waypoint: wait; if it persists, approach as close as clearance allows and hold there — a lower accuracy grade beats a missed stop.

### 9.5 Turning near ramps

The concept documents a ramp on the circuit (lip height and gradient unmeasured). Turning on a slope corrupts the heading (Eq. (20)). Sequence: (1) approach perpendicular to the sidewalk axis; (2) drive until all four wheels are on the flat; (3) stop; (4) ZUPT and pitch check; (5) turn in place by 90°; (6) drive 20 cm to 30 cm straight to align the casters; (7) re-acquire the edge. If the sidewalk at the ramp entry is narrower than the 0.6 m turning envelope, the turn happens on the slope: slope projection becomes mandatory and the turn is split into two 45° steps **[MEASURE]**.

### 9.6 Obstacle handling as a one-dimensional problem

The vehicle drives in a corridor whose boundaries it measures. Evasion is therefore not 2-D planning: the single free variable is the lateral offset $y_{off}$ from the nominal path within the corridor.

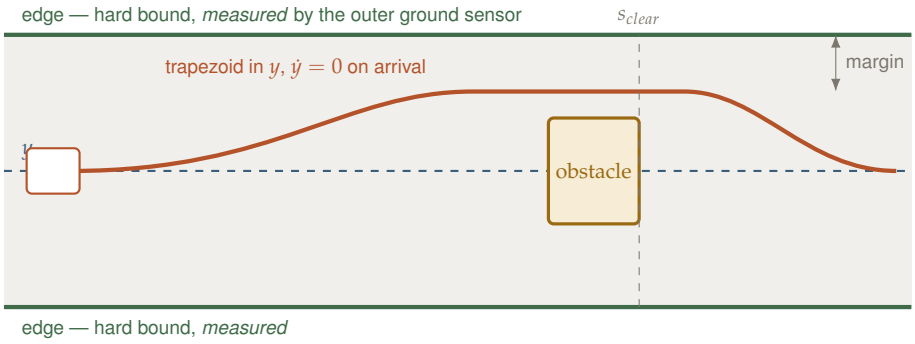



**Figure 10:** 1-D evasion. The free set of offsets is bounded by the measured edges (minus a safety margin) and by the obstacle's lateral extent. The commanded offset is the free value closest to $y_{nom}$, reached by a trapezoidal lateral profile with asymmetric limits. There is no separate return manoeuvre: once the vehicle passes $s_{clear}$, $y_{nom}$ is free again and the same rule returns to it.

- **Free set.** $\mathcal{Y} = [y_{min}, y_{max}] \setminus [y_{o,1}, y_{o,2}]$, with $y_{min,max}$ from the detected edges minus margin, and the obstacle interval from the forward sensors (ROI zones give a coarse profile of $\approx 6$ directions over $\approx 50°$).
- **Choice.** $y_{cmd} = \arg\min_{y \in \mathcal{Y}} |y - y_{nom}|$, held until $s > s_{clear}$ (a single scalar: the along-track coordinate beyond the obstacle plus vehicle length and margin).
- **Execution.** The offset is added to the path follower's reference; lateral speed and acceleration are limited (trapezoid in $y$), asymmetric so that moving out is faster than moving back.
- **Safety.** The inner ground sensors stay armed; if $\mathcal{Y}$ is empty the vehicle waits (temporary obstacle) and after a timeout reports `FAULT/RECOVER`.
- **Estimation.** Landmark and face fixes are suppressed while $|y_{off}| > 0.1$ m; P is *not* inflated by the manoeuvre itself, only when a reference is lost.

- **Grading.** Waiting earns 1 point, rerouting 2. A static obstacle on the path is the test case for week 16.

**Reflex layer.** Independent of all of this, the forward sensors stop the vehicle at 100 Hz without interpretation. At 0.4 m/s and ≈ 100 ms total latency the reaction distance is ≈4 cm; braking is latency-limited, not physics-limited. [RISK] Other vehicles carry the same 940 nm sensors: median filter over 3–5 frames, a plausibility gate on range jumps, and the signal rate consistency check reject phantom readings.

# 10 Communication

## 10.1 Links

| Link | MCU side | PC side | Rate | Use |
|---|---|---|---|---|
| USB console | USART2 (PA2/PA3) via ST-Link VCP | `/dev/cu.usbmodem...` | 115200 Bd 8N1 | development, calibration, high-rate logging over cable |
| Radio | USART1 (PB6/PB7) → HC-12 | HC-12 → FT232RL, `/dev/cu.usbserial-AQ8NFDGG` | 9600 Bd 8N1 | telemetry (week 10), remote control (week 14) |

Both consoles run in parallel with the same command set; a port is opened by one program at a time (`screen` *or* the CubeIDE terminal *or* a Python script).

## 10.2 HC-12 radio

- **Module.** SI4463 transceiver with an STM8 controller running vendor firmware; transparent UART; half-duplex. Channel $n$ at $f = 433.4 + (n-1) \cdot 0.4$ MHz; both modules on CH001, FU3, 9600 Bd. Transmit power P4 (+8 dBm) for bench tests, AT+P8 (+20 dBm) for the course.
- **Configuration.** SET low (PC9 on the vehicle), wait ≈ 40 ms, send the AT command in one piece (script, not `screen`), release SET, wait ≈ 80 ms. Baud, channel and FU mode must match on both ends; power need not.
- **Supply.** Up to ≈ 100 mA during transmit bursts, minimum 3.2 V. The FT232RL's 3.3 V output (≈ 50 mA) is insufficient — ground module on the FTDI's 5 V with the jumper set accordingly. On the vehicle the module overheated the AMS1117 once; see Section 3.3.
- **Channel plan.** Only CH001–CH004 lie inside 433.05–434.79 MHz; with the recommended channel spacing between pairs, only one pair fits there. Several teams cannot separate by channel alone: frame IDs and CRC are mandatory, and channels must be coordinated with the other teams [ASK].

**Regulatory status.** [RISK] Research in week 5 found the 433 MHz band allocated to the amateur service in Mexico rather than to licence-free use, and the HC-12 not legally usable in the 455 MHz to 463 MHz alternatives considered. A Digi XBee 3 Pro (2.4 GHz, IFT-certified) was identified as a compliant alternative; it contains its own microcontroller, to be disclosed under the "only STM32" criterion. [DECIDE] keep HC-12 (technically working) or move to a certified module; [ASK] the professor's position. [VERIFY] the regulatory finding against the current IFT tables before relying on it.

## 10.3 Present message formats

**Vehicle console (text).** Commands end with CR or LF: `D<mm>` drive distance, `V<mm/s>` nominal speed, `T<ms>[ <ms>]` ramps, `F/FL/FR` feedforward, `G<Kp> <Ki>` PI gains, `Y<K_sync>`, `k` parameter dump, `-100...100` open-loop duty, `s` stop, `r` reset counters, `p/a` print once/auto (250 ms). Run-time changes are lost on reset and must be copied into the `#defines`.

**Bench telemetry (CSV, 10 Hz).** `T,ms,mode,uL,uR,rpmL,rpmR,cntL,cntR,mV,flags`, integer-scaled ($u$ in 1/1000, rpm in 1/100). **Calibration firmware** prints machine-readable records `SWP`, `GND`, `BRK`, `BLS`, `ODO`, `FIT`, `SUM`.

**Air-time.** At 9600 Bd one byte takes 1.04 ms. A 60-character CSV line is 62.5 ms; at 10 Hz that occupies 62.5 % of the channel — and the HC-12 is half-duplex, so uplink commands collide with telemetry. A 24-byte binary frame needs 25 ms (25 % at 10 Hz).

### 10.4 Planned protocol

One binary protocol for both transports (cable and radio):

- **Framing:** COBS with a zero delimiter; header (vehicle ID, sequence number, message type), payload, CRC-16.
- **Vehicle is timing master.** It sends telemetry in fixed slots and leaves a gap for the uplink after each frame; the PC answers only in the gap. This removes half-duplex collisions without a handshake.
- **Idempotent absolute commands.** `DRIVE(`$v, \omega$`)` carries the absolute set-point, never increments; a lost or duplicated frame does no harm.
- **Messages:** `HELLO` (firmware Git hash), `TELEM_FAST` (pose, speeds, duties, mode, flags), `TELEM_SLOW` (battery, temperatures, covariance trace, counters), `DRIVE`, `MODE`, `PARAM_GET/SET`, `CAPTURE`, `ACK`.
- **Failsafe by mode:** MANUAL — no valid `DRIVE` for 500 ms $\Rightarrow v = \omega = 0$; AUTO — link loss does not stop the vehicle (autonomy must not depend on the link), but a received `STOP` always does.
- **CAPTURE.** Radio rate is enough to monitor, not to tune. On command the vehicle records 24-byte samples at 100 Hz into RAM for 20 s (48 KB of the 128 KB) and transmits them afterwards.
- **Link map.** Sequence gaps logged together with the pose map radio holes along the course; heartbeat timeouts are then chosen from data.

**Ground station.** Python 3 with pyserial in three layers: raw byte logger (always on, never loses data), live view, offline analysis; `calib_host.py` already implements logging, fitting and plotting for the calibration firmware. `[ASK]` whether the PC may receive telemetry during graded AUTO runs (passive reception is not an external aid to the sensors, but this should be confirmed).

### 10.5 UART transmit discipline

**Two alternating TX buffers are mandatory.** `HAL_UART_Transmit_IT` returns `gState` to READY when the last byte has been written to the data register, not when it has left the shift register. A single buffer reused at that moment is overwritten during the last character time (87 µs at 115200 Bd, 1.04 ms at 9600 Bd) — seen on the bench as characters missing at random positions, which looks like a baud-rate or noise problem and is neither. Console output is formatted with `vsnprintf` into a ring buffer (drop on full, never block); `printf` with floats is not used (+10–15 KB flash, and a blocking 80-character transmission at 115200 Bd takes 7 ms).

# 11 Software architecture and toolchain

## 11.1 Cyclic executive

No RTOS: one 1 kHz timer interrupt with counters drives all rates (harmonic nesting, so slots never collide); the ISR only takes snapshots and sets flags, the work runs in the main loop. No task ever blocks; all I/O is DMA or interrupt-driven.

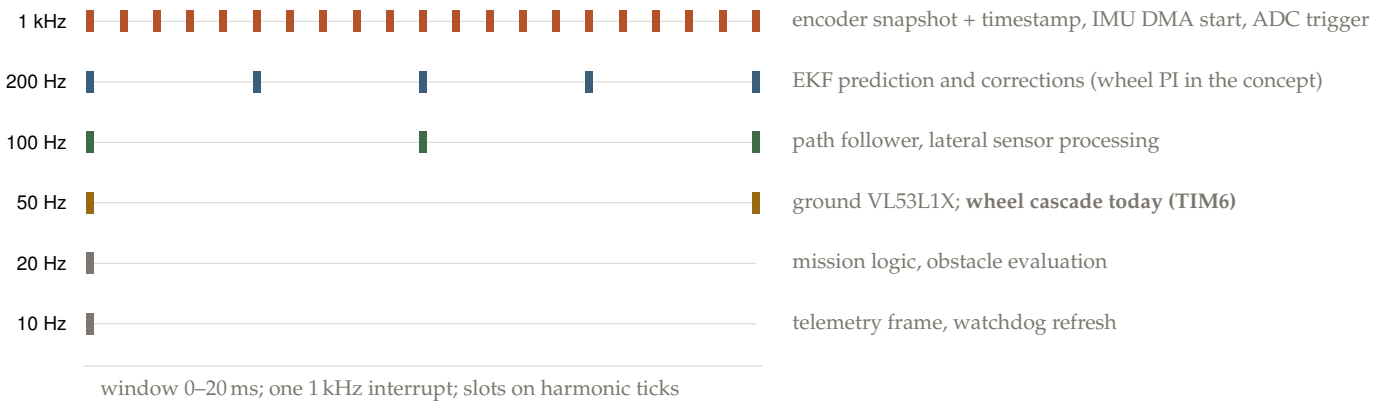



**Figure 11:** Target schedule. Every slot's execution time is measured with a GPIO toggle and an oscilloscope — the CPU load meter costs one pin.

**Today** the vehicle firmware runs the wheel cascade directly in the TIM6 interrupt at 50 Hz; the main loop parses the two consoles. The transition to the cyclic executive happens when the IMU and EKF arrive (week 8–10), with the wheel cascade kept at 50 Hz or moved to 200 Hz together with an M/T speed estimator (Section 14.7).

**Time base.** TIM2 free-running at 1 MHz (32 bit, wraps after 71.6 min); every measurement is stamped from it at acquisition, not from `HAL_GetTick()` (1 ms resolution, interrupt-dependent). The calibration firmware uses the DWT cycle counter for the same purpose and is therefore independent of the TIM6 rate (tested at 50 and 83 Hz with identical results).

## 11.2 Firmware components

| Firmware | Files | Content |
|---|---|---|
| Bench (week 5) | `bench_config.h`, `motor`, `encoder`, `bench` | open-loop drivetrain, console, test sequences 0–6 (truth table, sign, deadband, $K_v$ sweep, demos, hand-turn) |
| Vehicle (teammate, extended) | `main.c` (USER CODE blocks) | BTS7960, encoders, trapezoid + P + FF/PI cascade, L/R sync, `D<mm>`, HC-12 console |
| Calibration | `calib.c/.h`, `calib_config.h`, `calib_host.py`, `sim/` | sign, breakaway, sweeps, backlash, ground sweep, odometry runs (`D`, `W`, `Q`); selected by `APP_CALIB`; verified in software-in-the-loop simulation; 21–30 KB |

Module prefixes for the full firmware: `mot_`, `enc_`, `imu_`, `vl_`, `ekf_`, `nav_`, `msn_`, `com_`, `sys_`; units in identifiers (`_mm`, `_mms`, `_rad`).

## 11.3 Language and build rules

- C (not C++). Single precision only: every literal has `f`, no `double`, `-Wdouble-promotion` on (it is missing in the current project — O-11), `-ffast-math` forbidden (it changes IEEE semantics the EKF relies on). Console formatting with integers only.

- Development at `-Og` (current project: `-O0`, O-11); the full competition build at `-O2` in week 13, not 16 — every missing `volatile` surfaces on that day.
- Flash budget: 60–90 KB of 512 KB expected; VL53L1X Ultra Lite Driver ($< 10$ KB) instead of the full API ($\approx 90$ KB); `-Wl,-print-memory-usage` after every build.

### 11.4 CubeMX regeneration protocol

CubeMX owns the `.ioc`; code lives only in USER CODE blocks. Before every regeneration: commit. After: `git diff` of `main.c` and `.cproject`. CubeMX is known to reset values silently (TIM6 period, build flags). The clock tree (PLLM = 4, HSE BYPASS) and TIM6 period are checked in the generated code every time. Pins are reserved in `TOOLCHAIN.md` before they are touched in CubeMX (USART1 on PB6/PB7 must be assigned before TIM1 CH2/CH3, or CubeMX places it on PA9/PA10).

## 12 Error budget and expected accuracy

Worked example: a 40 m segment without external reference before a waypoint, driven at 0.4 m/s (100 s). Each term is computed with the model of Sections 4.8 to 4.10.

| Term | Assumption | Error at 40 m | Model |
|---|---|---|---|
| Entry heading error | 0.3° at segment start | 21 cm lateral | $s\,\delta\psi_0$ |
| after a face fit | 0.13° (VL53L1X, $\sigma_r = 2$ cm, 2 m, $N = 250$) | 8.8 cm lateral | Eq. (22) |
| Residual gyro bias | $1 \times 10^{-3}$ °/s after ZUPT | 3.5 cm lateral | $\delta b\, s^2/(2v)$ |
| Scale error | 0.5 % | 20 cm along | $\varepsilon_s s$ |
| after per-surface calibration | 0.2 % (`D5000`×3) | 8 cm along | |
| after a landmark fix near the waypoint | $\sigma \approx 3$ cm | 3 cm along | |
| Turn on a 6° slope (avoidable) | integrated body rate | 35 cm lateral | Eq. (20) |
| Backlash, one reversal | 1–2 mm and 0.3°–0.5° per reversal | avoid reversals | Section 4.3 |
| Stop accuracy of the controller | $\lvert e_x \rvert < 2$ mm | 0.2 cm | |

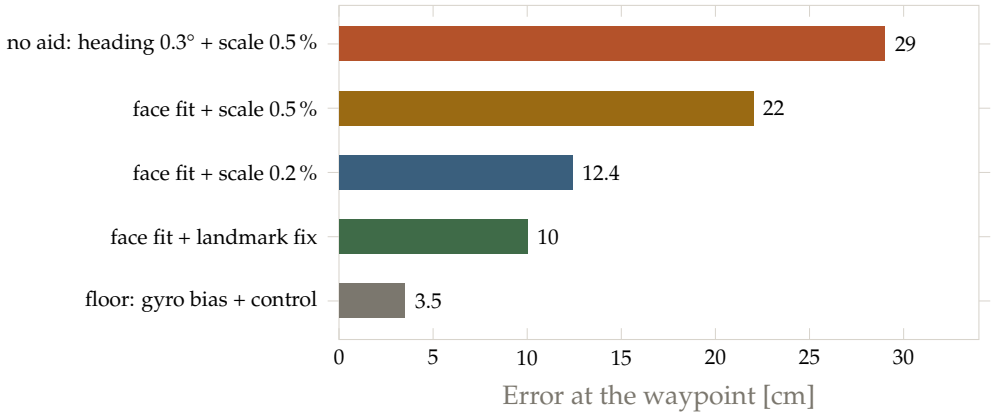



**Figure 12:** Root-sum-square of lateral and along-track terms (first bar $\sqrt{21^2 + 20^2 + 3.5^2} = 29$ cm). **The entry heading dominates the lateral error and the scale dominates the along-track error.** One face fit and one per-surface scale calibration move the waypoint into the 10 cm to 20 cm bracket (1.5 points); a landmark fix shortly before the waypoint reaches the 1.75 boundary. The 0 cm to 5 cm bracket needs both fixes *and* short reference-free stretches.

**Target and its justification.** Every waypoint inside 10 cm to 20 cm with a completed circuit. The two cheapest levers are the survey (landmarks and faces near each waypoint) and per-surface scale calibration; neither needs new hardware.

## 13 Calibration and verification plan

### 13.1 Calibration procedure (calibration firmware)

0. **Preparation.** Tyres clean, battery charged and mounted where it rides on the course (load distribution changes the rolling radius), battery voltage at the driver input measured and

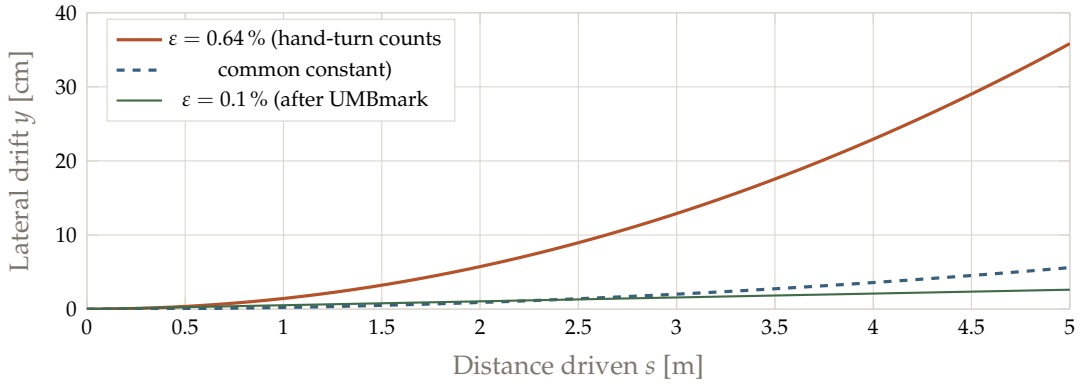



**Figure 13:** Encoder-only straight driving: a wheel-scale mismatch grows quadratically (Eq. (18)), a heading offset linearly. This is the behaviour of today's firmware (no gyro); with gyro heading the quadratic term leaves the estimate but remains in the L/R synchronisation.

entered (`U11.92`).

1. **Stand,** `A` (≈ 7 min): sign check, breakaway and stop duty (6× per direction), descending duty sweep (16 levels, 2×) giving $K_v$, $K_s$ per wheel and direction with standard errors, $R^2$, residuals and $\tau$ at every level change. Optionally at full and at low pack voltage.
2. **Backlash,** `L` (vehicle on the box, each tyre held by hand in turn): lost motion with and without torque.
3. **Ground,** `G` (≈ 1.2 m of floor): closed-loop sweep at 8 speeds forward and back — $K_v$, $K_s$ **under load**, the values for the vehicle firmware.
4. **Odometry** on course-type paving: (a) scale: `D5000` ×3 per surface, distance measured with tape; (b) UMBmark: `Q2000` ×5 and `Q-2000` ×5, each preceded by `D50` (gearboxes loaded forward) and marking the start; (c) optional wheelbase check `W3600/W-3600` (10 turns); (d) `odo` computes new mm/count L/R and wheelbase; enter with `O...`, repeat two squares each way — the return error should fall to a few millimetres.

**Pose measurement.** Two pointers on the centre line, one ahead of ($p_f = 150$ mm) and one behind ($p_r = 100$ mm) the axle; mark both on masking tape before and after a run, measure each displacement along and across the start axis with a carpenter's square. The two points give axle position and true heading change; errors are always *real minus encoder*, so the result does not depend on the controller.

### 13.2 UMBmark

Borenstein's UMBmark drives an $L \times L$ square clockwise and counter-clockwise; the centres of gravity of the return errors $x_{cw}, x_{ccw}$ separate the two systematic errors:

$$\alpha = \frac{x_{cw} + x_{ccw}}{-4L}, \quad \beta = \frac{x_{cw} - x_{ccw}}{-4L}, \quad R = \frac{L/2}{\sin(\beta/2)}, \quad E_d = \frac{R + b/2}{R - b/2}, \quad E_b = \frac{\pi/2}{\pi/2 - \alpha}, \tag{34}$$

($\alpha, \beta$ in rad, signs per Borenstein's convention), corrected by $b' = E_b\, b$, $D'_L = \frac{2}{E_d+1} D$, $D'_R = \frac{2}{1/E_d+1} D$. The calibration script does not use these small-angle closed forms but fits $E_d$ and $E_b$ with an exact arc model of the commanded square (Gauss–Newton over both directions and both coordinates); in simulation with known truth the return error fell from 250 mm to 400 mm to below 1 mm. UMBmark does not see the mean scale — hence step (a) first.

### 13.3 Verification by layer

| Layer | Acceptance test | State |
|---|---|---|
| Actuation | signs (`C`), no forbidden state, enables low in reset | passed |
| Speed loop | step 0→200 mm/s on the floor: overshoot $< 5\,\%$, no stick-slip at 0.1 m/s | to repeat on floor |
| Position loop | `D300`, `D1000`, `D5000`: end within 2 mm (encoder), no reversal, L/R difference $< 3$ mm | D1000 passed |
| Odometry | UMBmark return error $< 10$ mm for $L = 2$ m | open |
| IMU | Allan variance; bias vs temperature over 20 K | after selection |
| EKF stationary | bias converges in INIT; $P$ stays symmetric/positive | week 10 |
| EKF hand-pushed | 10 m straight: estimate within 10 cm | week 10 |
| Ground sensors | classify path/edging/grass on site, noon and afternoon | after fitting |
| Path follower | closes a 10 m × 10 m rectangle within 30 cm | week 13 |
| Mission | full circuit, waypoints within 40 cm (wk 15), 20 cm (wk 16) | weeks 15–16 |
| Radio | 15 min logging at 10 Hz, $< 1\,\%$ lost frames; failsafe on link loss | weeks 10/14 |

# 14 Open items, missing values and optimisation

Everything that is not yet measured, decided, implemented or verified is listed here once, with the sections it blocks. Priorities: blocking, high, medium, low.

## 14.1 Field measurements — nothing measured yet

| # | Measurement | Blocks | Prio |
|---|---|---|---|
| F-1 | Ramp-lip height | wheel and caster diameter (Section 4.6) | blocking |
| F-2 | Ramp gradient | motor margin, whether slope projection is mandatory | high |
| F-3 | Sidewalk width at the ramp entry | turning on level ground (Section 9) | high |
| F-4 | Rubber-edging height, constancy, flush or proud | ground-sensor thresholds | high |
| F-5 | Narrowest path width | maximum vehicle width | high |
| F-6 | Residual width in furniture zones | whether evasion is possible there | medium |
| F-7 | **Length of reference-free stretches, especially before each waypoint** | error budget — the single most important number | blocking |
| F-8 | Benches/tables: position ±5 cm, face orientation ±0.5° | heading fits, landmark map | blocking |
| F-9 | Posts: position and diameter | landmark map | high |
| F-10 | Waste bins and other second-class landmarks | landmark map | medium |
| F-11 | Total circuit length, leg lengths | time budget, path definition | high |
| F-12 | Leg headings | path definition | high |
| F-13 | Step: height, position, whether it is crossed anywhere | ground sensors, safety | high |
| F-14 | Surface types per leg (paving, concrete, edging) | per-surface scale calibration | medium |
| F-15 | Radio coverage along the course | heartbeat timeouts | low |

Method: tape measure and carpenter's square for widths and heights; a 30 m tape and pairwise distances (trilateration) between fixed points for the landmark map; photographs of every landmark from the path with a scale. One afternoon unblocks F-1 to F-14.

## 14.2 Vehicle parameters still to determine

| Parameter | Current value | Status | How |
|---|---|---|---|
| Vehicle mass $m$, load split axle/casters | — | [MEASURE] | scale under each wheel |
| $K_v$, $K_s$ under load, per wheel and direction | bench / wheels-in-air | [MEASURE] | `G` |
| Breakaway $K_{s,break}$ on BTS7960 | 14.1/16.9 % (bench) | [MEASURE] | `B` |
| Plant time constant on the floor | 50 ms (in air) | [MEASURE] | `G`, steps |
| Driver output floor $U_{min}$ at 20 kHz | — | [MEASURE] | oscilloscope |
| Armature $L_a$, $R_a$; $k_e$ | estimates only | [VERIFY] | LCR meter, locked-rotor test |
| Mean scale $D/C$ per surface | 72.6 mm/3584 | [VERIFY] | `D5000`×3 |
| Per-wheel mm/count ($E_d$) | equal | [MEASURE] | UMBmark |
| Effective wheelbase ($E_b$) | 223.25 mm geometric | [MEASURE] | UMBmark, `W±3600` |
| Backlash $\delta_b$ per wheel | — | [MEASURE] | `L` |
| Battery divider ratio | 5.7 nominal | [MEASURE] | multimeter vs ADC |
| Session current / energy | — | [MEASURE] | log $U_{bat}$, coulomb count |
| Gyro: ARW, bias instability, temperature coefficient | — | [MEASURE] | Allan variance, temperature run |
| VL53L1X: $\sigma_r$ outdoors, class thresholds, max range in sun | — | [MEASURE] | outdoor afternoon |
| EKF noise: $k_d$, $q_b$, $q_s$, $R$ per measurement | — | [MEASURE] | from the above |
| Path-follower gains $\omega_n$, $\zeta$ | design 1.5–2 rad/s, 0.7 | [VERIFY] | rectangle test |
| Sync gain $K_{sync}$ | 3 (mm/s)/mm | [VERIFY] | L/R difference on `D5000` |

## 14.3 Firmware and configuration items

Numbering follows the sheet “Offene Punkte” of the CubeMX workbook.

| # | Item | Why / how | Prio |
|---|---|---|---|
| O-1 | TIM6 period 199 | 119 gives 83 Hz against `CTRL_PERIOD_S` = 0.02; add a start-up printout of the measured rate | blocking |
| O-2 | End of move with $K_s$ intercept | wheel may stop short (static friction); breakaway switching or higher $K_i$; verify `D300/D1000` within 2 mm | high |
| O-3 | Battery operation without USB | HSE BYPASS depends on ST-Link MCO (Section 3.2.1); fallback crystal at X3 | high |
| O-4 | Hot AMS1117 | measure its input, supply HC-12 separately | high |
| O-5 | `V_NOM_MAX` 800 → 500 | drive saturates above ≈ 650 mm/s | medium |
| O-6 | `EVT_TIMEOUT` | settle timeout currently reports DONE | medium |
| O-7 | Per-wheel mm/count | L/R from UMBmark; raise $D$ to ≈ 73.0 mm if per-wheel counts are used | medium |
| O-8 | Straight-line drift | `D1000/2000/4000` lateral offset (shape tells the cause) | medium |
| O-9 | Speed limit under load | `D3000` at 300–500 mm/s, full and empty pack; 80 % of the first failing speed | medium |
| O-10 | HSE start-up timeout 1000 ms | RCC parameter; avoids failed start at power-on | medium |
| O-11 | Build flags | `-Og`, `-Wdouble-promotion`; `git diff .cproject` after regeneration | medium |
| O-12 | Error_Handler stops motors | `GPIOB->BSRR = (PIN12\|PIN13)«16`, CCR1–4 = 0 | high |
| O-13 | HC-12 channel coordination | all teams start on CH001 | medium |
| O-14 | $K_v$ sweep on the BTS7960 | superseded by calibration `A/G` | low |
| O-15 | PB7 pull-up, PH1 free | check `HAL_UART_MspInit`, pinout view | low |
| O-16 | CSS with callback | EN_L/EN_R low on clock failure | later |
| O-17 | TIM2 as 1 MHz time base | PSC 89, start base + IC | week 8 |
| O-18 | SPI1, I2C1/2, ADC1, DMA, IWDG | sensor bring-up | week 8+ |
| O-19 | Confirm which build is flashed | PI gains 0.10/0.50 vs SIMC 0.045/0.898; KS_BREAK switching | high |
| O-20 | Replacement Nucleo | board failure reported 10 Oct; re-verify clock tree on the new board | blocking |

## 14.4 Decisions

| # | Decision | Options and recommendation | Blocks |
|---|---|---|---|
| D-1 | IMU | ISM330DHCX (Adafruit) — decided | week-8 milestone |
| D-2 | Radio | keep HC-12 / XBee 3 Pro (certified, own MCU) | week 10/14 |
| D-3 | Caster type and size | after F-1 | chassis |
| D-4 | Ground-sensor mounting height and offsets | 100–200 mm, offsets 15/25 cm | spot size, thresholds |
| D-5 | HC-12 supply | 5 V branch (recommended) or second LDO | O-4 |
| D-6 | Hardware E-stop in series with EN | recommended, firmware-independent | safety |
| D-7 | PWM frequency | keep 20 kHz unless $U_{min}$ is large | low-speed resolution |
| D-8 | Wheel loop rate | 50 Hz now; 200 Hz only with an M/T estimator | path bandwidth |
| D-9 | Start-heading jig | physical alignment of $\{W\}$ | EKF init |
| D-10 | Spare parts | Nucleo, BTS7960, motor | schedule risk |
| D-11 | Second battery | decided; order | 3-h sessions |

## 14.5 Questions

| # | Question | To | Urgency |
|---|---|---|---|
| Q-1 | Is 3.4 kgf cm the rated or the stall torque? (model says rated) | vendor | low |
| Q-2 | Restock lead time HS5141-178 (spare) | vendor | medium |
| Q-5 | Do two driven wheels plus two casters satisfy “four-wheeled”? | professor | highest |
| Q-6 | How many vehicles on the course simultaneously? | professor | high |
| Q-7 | Furniture in the same place on exam day? | professor | high |
| Q-8 | Legal radio band in Mexico for the link | IFT / professor | high |
| Q-9 | HC-12 FU mode and over-the-air rate | datasheet | low |
| Q-10 | Are radio modules and ToF sensors with their own firmware acceptable under “only STM32”? | professor | high |
| Q-11 | May the PC receive telemetry during graded AUTO runs? | professor | medium |
| Q-12 | How is waypoint accuracy aggregated (per waypoint, mean, worst)? | professor | medium |

(Q-3 connector pinout and Q-4 driver truth table are answered; Section 5.1.)

## 14.6 Risk register

| Risk | Impact | Mitigation |
|---|---|---|
| Course not surveyed in time | wheel size, landmarks, budget all unknown | one afternoon, now (F-1…F-14) |
| No spare parts | a failure costs a week (already happened once) | D-10 |
| IMU not delivered by week 8 | milestone and all heading aiding late | order the ISM330DHCX now; placeholder IMU for bring-up |
| Clock without USB fails | no autonomous run on battery | O-3: test, power bank, crystal |
| CubeMX regeneration reverts values | wrong tick rate, lost flags; “mysterious” behaviour | commit, `git diff`, start-up rate printout |
| Turn on a slope | 35–62 cm cross-track error from one manoeuvre | turn on level ground, slope projection |
| Ground sensors in sun | missed edge ⇒ grass ⇒ invalid run | thresholds tested at noon; inner interlock |
| Other robots’ ToF/radio | phantom ranges, collisions on CH001 | median + gate + signal check; frame IDs + CRC; channel plan |
| Single TX buffer race | corrupted frames that look like noise | double buffering (done in bench firmware) |
| Safety rule vs measurement | supervisor aborts its own test | scope stall checks per mode |
| Motor pairing (12 %/20 %) | arcs in open loop | per-wheel FF + PI + sync (done) |
| Dupont on motor leads | heat, drifting series resistance | crimped/soldered ≥20 AWG before driving |
| Regulatory status of 433 MHz | module unusable officially | D-2, Q-8 |

## 14.7 Optimisation roadmap

Ordered by expected points per effort. Items 1–5 need no new hardware.

1. **Survey the course** (F-1…F-14). Unblocks everything below and defines where the heading fits and landmark fixes come from.
2. **Per-surface scale and UMBmark.** Per-wheel mm/count, effective wheelbase, and backlash as a per-wheel dead band in odometry (first $\delta_b$ counts after a reversal do not move the wheel).

Expected: return error of a 2-m square from centimetres to millimetres; along-track scale to 0.2 %.

3. **Battery-normalised feedforward.** Fit the feedforward in volts ($K_v^V$, $K_s^V$ with $U_{bat}$ recorded), measure $U_{bat}$ on PA0 at 20 Hz with a low-pass, apply $d_{ff} = (\ldots)/U_{bat}$. Removes a 27 % gain variation over the discharge from the PI's job.
4. **Ground-identified constants** from `G` and `B`; the breakaway switching with measured values; inertia feedforward with the floor $\tau_m$.
5. **Better low-speed speed estimate (M/T method).** Sample the encoder counter with a 1 µs timestamp at 1 kHz and compute speed over the interval between the first and last *count change* inside a window, $\hat{v} = c\,\Delta n/\Delta t_{edges}$. Resolution then improves from 3.18 mm/s to well below 1 mm/s at approach speeds, the IIR filter (29 ms lag) can be lighter, $\theta$ shrinks, and SIMC allows a faster speed loop — which in turn allows a faster position and path loop.
6. **Grade feedforward.** With pitch $\vartheta$ from the IMU, add the duty that holds the vehicle on the slope, $\Delta d \approx \frac{R_a}{k_t U_{bat}} \cdot \frac{mg \sin\vartheta\, r}{2\, i\, \eta}$, identified directly as $\Delta d / \sin\vartheta$ from a ramp test. The PI then does not have to wind up on the ramp.
7. **ZUPT at every stop**, not only at waypoints — in particular before every turn (fresh bias immediately before the manoeuvre that converts heading error into cross-track error).
8. **Gyro temperature model** from a warm-up run; 10-minute warm-up before graded runs.
9. **Heading fits at every straight face** before reference-free stretches, and a landmark fix within a few metres of each waypoint (Fig. 12).
10. **Path-follower gain scheduling** ($k_y = \omega_n^2/v$) and curvature feedforward on ARC segments; saturation that keeps $\omega$ and reduces $v$.
11. **Offline optimisation from practice runs.** Log full-rate data with `CAPTURE`; re-estimate scale, wheelbase, sensor offsets and the landmark map by smoothing the recorded runs; burn the constants in (week 15).
12. **Safety hardening:** hardware E-stop in the EN lines, IWDG, CSS with motor stop, Error_Handler that stops the motors.
13. **Mechanics:** hard, narrow tyres (stable rolling radius, defined contact point and therefore defined $b$); IMU rigidly at the axle midpoint; battery fixed where it rides during calibration.

**Next five actions.** (1) Fix TIM6 = 199 and confirm the flashed build (O-1, O-19) on the replacement board (O-20). (2) Survey the course (F-1…F-14). (3) Run calibration `A`, `L`, `G` and the odometry sequence. (4) Order the IMU (D-1, ISM330DHCX); the five VL53L1X are ordered. (5) Ask the professor Q-5, Q-7, Q-10.

# A Constants register

All numerical constants of the vehicle firmware (`main.c`, USER CODE PD) and the physical constants used in this document. *Source:* M = measured, C = catalogue, D = design value, E = estimate.

| Constant | Value | Unit | Src | Note |
|---|---|---|---|---|
| **Mechanics and odometry** | | | | |
| `WHEEL_DIAMETER_MM` | 72.6 | mm | M | effective, valid with $C = 3584$ |
| `COUNTS_PER_WHEEL_REV` | 3584 | counts | D | $64 \times 56$; measured 3595/3618 |
| `MM_PER_COUNT` | 0.063638 | mm | M | ratio $\pi D / C$ |
| `MM_PER_COUNT_L/R` | $\times 1.0000$ | — | D | per-wheel factors, from UMBmark |
| `WHEEL_BASE_MM` | 223.25 | mm | M | geometric; effective from UMBmark |
| **Signs and actuation** | | | | |
| `ENC_L_SIGN / ENC_R_SIGN` | $+1$ / $-1$ | — | M | vehicle harness |
| `MOT_L_SIGN / MOT_R_SIGN` | $-1$ / $+1$ | — | M | positive duty $\Rightarrow$ forward |
| `PWM_FREQ_HZ` | 20000 | Hz | D | TIM1 ARR 8999 |
| **Control loop** | | | | |
| `CTRL_PERIOD_S` | 0.02 | s | D | must equal the TIM6 period (199) |
| `SPEED_FILTER_ALPHA` | 0.5 | — | D | IIR, $\tau_f = 29$ ms |
| `KV_L / KV_R` | 0.00492 / 0.00507 | duty/rpm | M | bench |
| `KS_L / KS_R` | 0.110 / 0.127 | duty | M | bench intercepts |
| `FF_KS_L/R_PCT` | 9.8 / 4.1 | % | M | BTS7960 in air (revised build) |
| `FF_KV_L/R` | 0.1275 / 0.1361 | %/(mm/s) | M | BTS7960 in air (revised build) |
| `KS_BREAK_L/R_PCT` | 14.1 / 16.9 | % | M | bench breakaway; replace by B |
| `TAU_PLANT_S` | 0.050 | s | M | inertia feedforward |
| `V_MOVING / V_STOPPED / V_ZERO` | 8 / 3 / 2 | mm/s | D | friction switching, zero command |
| `KICK_DUTY_PCT / KICK_TICKS` | 40 / 5 | %, ticks | D | superseded by breakaway switching |
| `KP_SPEED / KI_SPEED` | 0.10 / 0.50 $\rightarrow$ 0.045 / 0.898 | %/(mm/s), /s | D | SIMC balanced |
| `INTEG_MAX_PCT` | 30 | % | D | anti-windup |
| `KP_POS` | 2 | (mm/s)/mm | D | $\tau = 0.5$ s |
| `V_CORR_MAX_MM_S` | 100 | mm/s | D | position-loop clamp |
| $K_{sync}$ (Y) | 3.0 | (mm/s)/mm | D | L/R synchronisation |
| `SETTLE_V_MM_S / POS_TOL_MM` | 20 / 2 | mm/s, mm | D | end of move |
| `SETTLE_TICKS / FAULT_TICKS` | 50 / 30 | ticks | D | 1 s / 0.6 s |
| **Limits and interface** | | | | |
| `V_NOM_DEFAULT / V_NOM_MAX` | 300 / 800 | mm/s | D | max to 500 (O-5) |
| `ACC_MS / DEC_MS_DEFAULT` | 1000 / 1000 | ms | D | 0.3 m/s$^2$ at 300 mm/s |
| `MIN / MAX_DIST_MM` | 5 / 20000 | mm | D | |
| `DEADMAN_MS` | 2000 | ms | D | open loop only |
| `AUTO_PRINT_MS` | 250 | ms | D | |
| `RX_LINE_MAX` | 16 | chars | D | |
| **Physical and electrical** | | | | |

| Constant | Value | Unit | Src | Note |
|---|---|---|---|---|
| Gear ratio | 56.3 | — | C | implied 56.17 / 56.53 (hand-turn) |
| Counts per motor revolution | 64 | — | M | 16 PPR ×4 |
| No-load speed | 181 / 177 | rpm | M | bench, $d = 1$ |
| Stall current | 4.3 | A | C | per motor |
| Rated torque | 3.4 | kgf cm | C | rated vs stall: Q-1 |
| $k_e$, $R_a$ | 10.0 mV s/rad, 2.8 Ω | — | E | Section 4.4 |
| Plant $K$, $\tau_m$, $u_0$ | 7.596, 50 ms, 7.0 % | — | M | BTS7960, in air |
| Battery | 9.9–12.6 | V | C | 3S, 2.2 Ah, 35C |
| Cut-off | 10.0 | V | D | |
| Divider | 10/57 | — | D | 47 kΩ/10 kΩ |
| Encoder pull-ups | ≈ 3 | kΩ | M | A–B 6 kΩ |
| Encoder filter | 2.84 | µs | D | IC filter 15 |

# B Glossary

| Term | Meaning |
|---|---|
| ARR | auto-reload register of a timer: the counter period minus one |
| ARW | angle random walk: integrated white rate noise of a gyroscope, in $^\circ/\sqrt{\text{h}}$ |
| CCR | capture/compare register of a timer channel; sets the PWM duty |
| CKD | clock-division field of a timer; sets the digital-filter sampling clock $f_{DTS}$ |
| COBS | consistent-overhead byte stuffing: framing that removes zero bytes so that 0 can delimit frames |
| CRC | cyclic redundancy check |
| CSS | clock security system: hardware fallback to HSI and NMI when the external clock fails |
| DMA | direct memory access: data transfer without CPU involvement |
| DRDY | data-ready output of a sensor |
| DWT | data watchpoint and trace unit; contains the CPU cycle counter |
| E5V | external 5 V supply input of the Nucleo (CN7-6) |
| EKF | extended Kalman filter: Kalman filter on linearised non-linear models |
| FF | feedforward: control action computed from the model and the set-point, not from the error |
| HSE / HSI / LSI | high-speed external / high-speed internal / low-speed internal clock |
| ICR | instantaneous centre of rotation |
| IIR | infinite-impulse-response (recursive) filter |
| IWDG | independent watchdog (own LSI clock) |
| MCO | microcontroller clock output; here of the ST-Link MCU, used as HSE |
| M/T method | speed measurement combining a count (M) with a measured time between edges (T) |
| NHC | non-holonomic constraint: zero lateral velocity of a wheeled vehicle |
| PI | proportional-integral controller |
| PLL | phase-locked loop; multiplies the input clock to the system clock |
| PPR | pulses per revolution (per channel) |
| PWM | pulse-width modulation |
| ROI | region of interest: active SPAD sub-array of the VL53L1X |
| SIMC | Skogestad internal model control: analytic PI/PID tuning rules |
| SPAD | single-photon avalanche diode |
| ToF | time of flight |
| UMBmark | University of Michigan Benchmark: square-path test separating wheel-diameter and wheelbase errors |

| Term | Meaning |
|---|---|
| VCP | virtual COM port of the ST-Link |
| XSHUT | shutdown input of the VL53L1X, used for address assignment |
| ZUPT | zero-velocity update: using standstill as a measurement (here of the gyro bias) |

# C References


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